Overcoming technological barriers for blockchain adoption in supply chains: a diffusion of innovation (DOI)-informed framework proposal

Katherine Kaneda Moraes
Faculty of Engineering, Federal University of Grande Dourados (UFGD), Dourados, Brazil and
Department of Production Engineering, Federal University of São Carlos (UFSCar), São Carlos, Brazil

Gilberto Miller Devós Ganga
Department of Production Engineering, Federal University of São Carlos (UFSCar), São Carlos, Brazil

Moacir Godinho Filho
Metis Lab, EM Normandie Business School, Normandie, France and
Department of Production Engineering, Federal University of São Carlos (UFSCar), São Carlos, Brazil

Luis Antonio Santa-Eulalia
Department of Information Systems and Quantitative Methods in Business, Business School, Université de Sherbrooke, Sherbrooke, Canada, and

Guilherme Luz Tortorella
Department of Mechanical Engineering, The University of Melbourne, Melbourne, Australia; IAE Business School, Universidad Austral, Buenos Aires, Argentina and Department of Systems and Production Engineering, Federal University of Santa Catarina (UFSC), Florianopolis, Brazil

Abstract

Purpose – The integration of blockchain technology (BT) in supply chain management (SCM) is at the forefront of technological advancements, yet it faces significant barriers that hinder its widespread adoption. This study aims to delve into these challenges, employing the diffusion of innovations (DOI) theory to systematically investigate and propose a strategic framework for overcoming the technological barriers to BT adoption within SCM.

Design/methodology/approach – Through a comprehensive systematic literature review (SLR) of 155 publications, complemented by rigorous content analysis and expert interviews, this research identifies and categorizes 16 primary technological barriers, including scalability and privacy issues, that impede BT integration.

Findings – The proposed framework, informed by DOI theory, outlines tailored strategies across three critical adoption stages: initiation, where the focus is on mitigating high energy consumption and scalability issues; adoption decision, emphasizing the formulating international standards for blockchain architecture, embedding abstraction layers within software projects; and implementation, concentrating on enhancing security, interoperability and system efficiency.

Originality/value – This research contributes significantly to both academic literature and practical applications. Academically, it extends the DOI theory within the SCM context and enriches the blockchain literature by providing a nuanced understanding of the specific barriers to BT adoption. Practically, it offers a roadmap for industry practitioners, delineating actionable strategies to navigate the adoption process effectively. This study not only bridges the gap between theoretical insights and practical implementations but also serves as a vital resource for policymakers and standard-setting bodies in facilitating and regulating BT adoption in SCM, thereby fostering innovation and competitive advantage in the marketplace.

Keywords Blockchain technology, Supply chain, Technological barriers, Diffusion of innovation theory, Adoption stages

Paper type Research paper

1. Introduction

Blockchain technology (BT) is a disruptive technology that promises to revolutionize traditional business models

The current issue and full text archive of this journal is available on Emerald Insight at: **https://www.emerald.com/insight/1359-8546.htm**

Supply Chain Management: An International Journal
30/1 (2025) 19–49
© Emerald Publishing Limited [ISSN 1359-8546]
[DOI 10.1108/SCM-02-2024-0138]

The authors would like to acknowledge Capes and CNPQ Brazilian Funding Agencies for partially funding this research.

Declaration of interest: The authors declare that there is no conflict of interest.

Data Availability Statement: The data that support the findings of this study are available from the corresponding author.

Received 27 February 2024
Revised 29 June 2024
30 July 2024
Accepted 31 July 2024

(Queiroz *et al.*, 2019a). The BT concept, introduced by Nakamoto (2008), is considered one of the essential technologies for the Fourth Industrial Revolution. Blockchain research and initiatives cover areas such as the agriculture and food sector (Li *et al.*, 2023; Zhao *et al.*, 2019), healthcare (Tandon *et al.*, 2020), supply chain and logistics (SC&L) (Wang *et al.*, 2019b) and sustainability management (Kouhizadeh *et al.*, 2021), among others. BT has become the new architecture for decentralized transactions, maintenance and data sharing, pushing the boundaries of financial applications (Ziolkowski *et al.*, 2020).

BT consists of a distributed and secure ledger, storing transaction data in blocks (Kamble *et al.*, 2019; Olsen and Tomlin, 2020). Each block is integrated with the previous block, which facilitates retrieving and tracking historical information by exploring the previous blocks (Queiroz and Fosso Wamba, 2019). Each block has its own hash, which works as a unique ID, and the transactions registered and validated are given a unique sequence and time (timestamp), which guarantees the immutability of the records (Makridakis and Christodoulou, 2019; Queiroz and Fosso Wamba, 2019). BT can change the paradigm of supply chain (SC) transactions (Kamble *et al.*, 2019), improving visibility and transparency and eliminating trust issues in SCs (Essien *et al.*, 2023; Wan *et al.*, 2020).

BT is a peer-to-peer transaction platform that does not need any third-party intermediary (Kamble *et al.*, 2019). This platform ensures free communication among the blockchain nodes, where the nodes are geographically dispersed but equally privileged participants in the application (Feng *et al.*, 2019). Blockchain comprises a set of blocks that store specific information and are linked with cryptographic protocols (Biswas and Gupta, 2019).

Regarding BT's operation mode, we resorted to the explanation provided by Saberi *et al.* (2019b). When an agent creates a new transaction to be added to the blockchain, it is transmitted to the network for verification and auditing. If most nodes in the chain approve this transaction, considering the prespecified approved rules, this new transaction is added to the network as a new block. Once the new block is added to the blockchain, several copies are created decentralized, aiming to create a chain of trust. These authors also mention the smart contract as BT's critical feature, allowing for credible transactions without third parties' involvement.

BT is different from other technologies that have been used in SC. Currently, SCs largely rely on centralized information management systems within organizations, such as enterprise resources planning (ERP), which has its pitfalls (Saberi *et al.*, 2019b). Since BT allows for the maintenance of data throughout a product's lifecycle as it moves through SC, the technology acts as a trustworthy record keeper, fostering data nonrepudiation and immutability (Azzi *et al.*, 2019; Olsen and Tomlin, 2020; Tripathi *et al.*, 2020). In addition, BT distinguishes itself from all other technologies by eliminating the necessity for intermediaries, thereby decentralizing commercial contact and achieving a mutual accord that is based on consensus (Prasad *et al.*, 2018; Rane and Narvel, 2019). BT can still offer undeniable proof of the data's original source. With BT, any good's origin can be traced back to its root, and information (e.g. ownership data, time stamping and location data) is stored as they pass between various parts along the SCs (Abeyratne and Monfared, 2016; Kamble *et al.*, 2019).

The adoption of BT in supply chain management (SCM) is markedly hindered by a lack of well-documented use cases, field experiences and clear insights into BT trial outcomes. This challenge is further compounded by the limited literature addressing the technological barriers to BT adoption, as highlighted by Hackius and Petersen (2020), Queiroz *et al.* (2019b) and Dutta *et al.* (2020). This occurs even though many companies are actively developing and testing various prototypes. For example, the recent announcement of the discontinuation of TradeLens operations, joint initiative by A.P. Moller – Maersk and IBM for SC digitization, reinforced that the successful implementation of BT implies overcoming obstacles that limit its adoption.

While our study primarily examines the barriers to the adoption of BT, it is important to emphasize that our objective is not to demonstrate how these barriers impede the adoption of BT or to argue against the viability of this technology. Our objective is to provide readers with information regarding the actual existence of these barriers and strategies to overcome them. Multiple recent research has documented a series of successful BT implementations (Adel and Younis, 2023; Alkafaji *et al.*, 2023; Bag *et al.*, 2023; Chittipaka *et al.*, 2023; Sumarliah *et al.*, 2023).

The availability of successful implementations is likely associated with the existence of strategies to overcome existing barriers. Increasing the amount of knowledge regarding current barriers, the specific point in the process of adopting BT when they occur and the strategies that can be used to reduce their impact, can significantly enhance the quality of literature on BT adoption. Therefore, a study that sheds light so that companies and researchers know what they are dealing with when they decide to adopt BT becomes essential to contribute to the continued development of BT applications in SCs.

Recognizing the critical need for a deeper understanding of these impediments, this study embarks on a systematic exploration of both the technological barriers to BT integration in SCM and the strategies identified in the literature to surmount these obstacles. Our research extends beyond merely cataloguing the advancements and general barriers to BT adoption. We aim to provide a detailed examination of the specific technological barriers within the SCM context and to organize the strategies found in the literature that address these barriers, using the diffusion of innovation (DOI) theory as a guiding framework. This comprehensive analysis is facilitated through a multifaceted approach that combines a systematic literature review (Tranfield *et al.*, 2003) with rigorous content analysis (Krippendorff, 2019) and insightful expert interviews. This triangulation method enriches our understanding of the barriers and enhances the robustness of the strategies we identify, ensuring they are grounded in theoretical and practical insights.

The objective of this study is to provide novel insights that can be applied in both practical and theoretical contexts. It does so by integrating disparate themes, such as barriers and mitigation strategies, with the stages of technology adoption. The main goal is to gain a more comprehensive understanding of the limitations that impede the adoption of BT and how to overcome them. In contrast to other literature reviews, our

study not only summarizes the barriers to BT adoption but also explores and discusses strategies to mitigate them. In addition, the application of a theoretical lens enables a more accurate contextualization and interpretation, as well as a more robust conceptual foundation for the SLR results within a particular structure, such as the stages of innovation adoption, as per DOI theory. Additionally, our review suggests a well-defined research agenda for the future.

The literature suggests that research be conducted examining the barriers to BT adoption in this field. Rogers (2003) states that the innovation-decision process involves uncertainty about the expected consequences, and to deal with this, the decision-maker seeks to answer questions such as: "What operational problems am I likely to encounter, and how can I solve them?". A theoretical view is necessary to understand better the underlying motivators and barriers that will lead SC managers or discourage them from adopting BT (Francisco and Swanson, 2018). In this regard, it is important to employ a consolidated theoretical lens that enables us to comprehend the phases of BT adoption more systematically. The DOI theory (Rogers, 2003) was implemented in this study. This theory views the adoption of technology as a process that is characterized by a sequence of steps that occur over time. Consequently, it enables us to identify the specific moment in the adoption process at which companies will face BT implementation barriers and when mitigation strategies will become necessary.

Future studies should examine barriers and success factors in the BT adoption process (Schmidt and Wagner, 2019). Queiroz *et al.* (2019b) and Dutta *et al.* (2020) recognize that a comprehensive examination of the challenges that limit the adoption of BT in SCM is a research gap. Our study intends to help better understand these underlying barriers, specifically the technological ones. Consequently, we formulated the subsequent Research Questions (RQs) to direct our investigation:

RQ1. What are the technological barriers faced by companies during the adoption and use of blockchain within their SC operations?

RQ2. What are the recommended strategies to mitigate these barriers?

RQ3. In the context of the DOI theory, at which specific stages of blockchain adoption do the identified barriers and their corresponding mitigation strategies emerge?

This study makes contributions in different ways: it describes a complete and exhaustive list of technological barriers to BT adoption in SCM, including privacy, scalability and security issues, lack of interoperability and immutability; It suggests a discussion regarding the strategies for overcoming technological barriers that emerge during the three critical stages of technology adoption, defined using the lens of DOI theory. Based on DOI theory, the decision to embrace a technology does not happen instantaneously but rather unfolds gradually over a period of time. DOI theory allows for the precise definition of the activities carried out at each stage of the technology adoption process. This provided us with the foundation for constructing our framework. In this manner, we offer a roadmap for industry practitioners to develop actionable strategies for successfully traversing the BT adoption process. This comprehensive analysis represents a valuable result for the SCM field, filling a considerable gap in the literature, especially by integrating these barriers and relevant solutions.

The paper is organized into five sections. Section 2 provides the theoretical background. Section 3 describes the research methodology. Section 4 presents the findings and discussion. Section 5 summarizes the conclusions of this research, academic and practical contributions and proposes a future research agenda.

2. Conceptual and theoretical background

2.1 Diffusion of innovation theory

Researchers study organizational adoption using various theories to discern the factors influencing innovation acceptance across different IT adoption contexts (Frambach and Schillewaert, 2002). There is no single theory of innovation, nor does it seem likely that one will emerge (Fichman, 2000). Adoption decisions occur at two levels: the organizational level (by the organization itself) and the individual level (within the organization) (Frambach and Schillewaert, 2002).

Among the most used theoretical models at the organizational level are the diffusion of innovation theory (DOI) (Rogers, 2003), the technology-organization-environment (TOE) framework (Tornatzky *et al.*, 1990) and, at the individual level, the theory of reasoned action (TRA) (Fishbein and Ajzen, 1977), the technology acceptance model (TAM) (Davis, 1989) and the theory of planned behavior (TPB) (Ajzen, 1991).

Blockchain constitutes a technological innovation (Swan, 2015). Thus, theories concerning IT innovations are applicable to studies on BT adoption. A similar association, i.e. between IT innovation adoption and technology adoption, was previously made by Hameed *et al.* (2012). Theories and models that address IT innovation adoption are being used in studies that investigate the adoption of BT in the SC context, such as DOI theory (Nath *et al.*, 2022), TOE framework (Clohessy and Acton, 2019), TAM (Queiroz and Fosso Wamba, 2019), among others.

Other BT studies used DOI lenses. A multistage adoption model for a better understanding of blockchain diffusion across SCs was proposed, based on DOI as one of its main theories (Queiroz *et al.*, 2019b). Agi and Jha (2022) used the lens of the DOI to identify enablers of blockchain adoption and develop a comprehensive framework for blockchain adoption in the SC. Rogers' classical model of diffusion represents the closest attempt to formulate a general theory for innovation adoption (Rogers, 2003). Consequently, our study will utilize the DOI as the primary theory to elucidate the adoption stages of BT.

According to the DOI theory, the innovation-decision process is the process by which the decision-maker goes through the stages of obtaining initial knowledge of an innovation, forming an attitude toward the innovation, deciding to adopt or reject, implementing the new idea and confirming this decision (Rogers, 2003). This process consists of five stages:

1 Knowledge;
2 Persuasion;
3 Decision;
4 Implementation; and
5 Confirmation.

In the Knowledge stage, the decision-making unit is exposed to innovation and understands how it works. Persuasion is the stage in which information about the new idea is actively sought, intending to form a favorable or unfavorable attitude toward it. In the decision stage, activities are carried out that will lead to a choice to adopt or reject the innovation. Implementation is the stage where the innovation is put to use. It involves an evident change in behavior, as the new idea is put into practice. At the confirmation stage, the aim is to reinforce an innovation decision already taken, and in conflicting cases, to discontinue due to replacement or disenchantment that may occur.

According to the DOI theory, "each stage in the innovation-decision process is a potential rejection point." This implies that a BT adoption project, being a technological innovation, can be rejected at any of these five stages. This occurred in the case of TradeLens. This platform operated effectively for 5 years, and after surviving the Knowledge, Persuasion, Decision, and Implementation stages, it was discontinued at the Confirmation stage due to collaboration challenges between SC partners.

Innovation adoption is deemed successful when the innovation is accepted, embraced, and integrated into the organization, and individuals consistently continue to use it over an extended duration, as stated by Gopalakrishnan and Damanpour (1997) and Upadhyay (2020). Given the possibility of BT adoption being rejected at any point in the innovation adoption process, it is necessary to conduct studies that shed light on the main challenges and barriers that could emerge at each of these stages. More importantly is to highlight the actions that firms implementing BT should take to overcome these barriers at the appropriate times so the initiative for BT adoption is successful and not interrupted. This demonstrates the significance of this research, which attempts to outline the barriers and strategies for overcoming them at each stage of the innovation adoption process. The DOI theory will be used as the foundation for these stages.

Hameed *et al.* (2012) proposed a conceptual model for IT innovation adoption. Despite referring to other models and theories, such as the TOE Framework, TPB, TAM and TRA, among others, the authors used the DOI as the main basis for defining three stages in the process of adopting innovation at the organizational level:

1 Initiation;
2 Adoption-decision; and
3 Implementation.

Upadhyay (2020), based on the conceptual model proposed by Hameed *et al.* (2012), created a framework that guides the BT adoption process by organizations. In this framework, Upadhyay (2020) lists the activities necessary for adopting BT within the three stages presented in the conceptual model by Hameed *et al.* (2012). In addition to the DOI theory, our study will be based on the adaptations to this theory made by Hameed *et al.* (2012) and Upadhyay (2020) for the proposal of our framework.

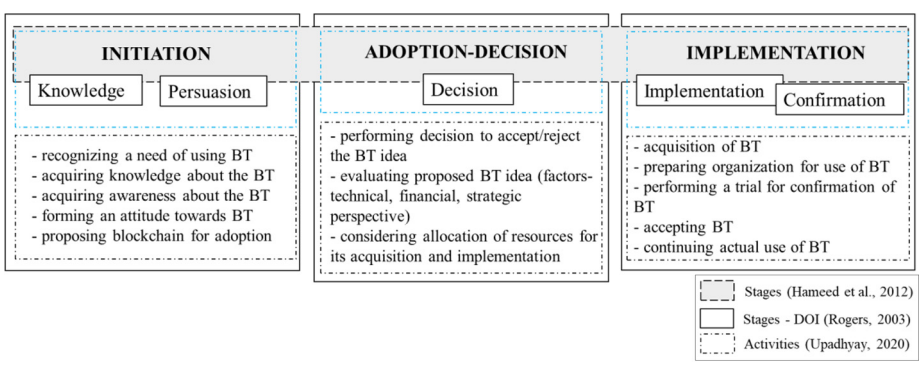
Figure 1 shows the relationship between the stages of DOI theory, the stages proposed by Hameed *et al.* (2012) and the activities proposed by Upadhyay (2020).

2.1.1 Initiation stage of adopting blockchain technology

Upadhyay (2020) states that during the initial phase of implementing BT, the organization becomes conscious of the necessity to utilize BT. This need may be driven by challenges or opportunities in its processes, which could be advantageous with the use of BT. For instance, the organization might have challenges related to trust, transparency or security that could be effectively addressed by utilizing the immutability features offered by BT (Kumar Singh *et al.*, 2023).

After identifying this need, firms strive to comprehend the underlying principles associated with BT. This entails comprehending the intricacies of data storage, validation, security mechanisms and the various types of blockchain, including public, private and hybrid. Once firms understand the technical aspects of BT, they seek to familiarize themselves with its real-world use cases across various industries. By doing a thorough investigation of use cases, the advantages of BT may be uncovered and demonstrated to its users (Behnke and Janssen, 2020).

Figure 1 Blockchain adoption-decision process



Source: Authors own work

Once firms have comprehended the technological intricacies and acquired a level of familiarity with BT, they may evaluate its suitability in addressing the challenges that they encounter and develop a positive or negative attitude toward the technology. At this initial stage, if the organization develops a positive attitude toward implementing BT, it may suggest its adoption. This step entails discerning the specific opportunities within the firm where the technology can be efficiently utilized. Specifically, the corporation is starting to develop a more concrete attitude toward technology in the context of its operations.

2.1.2 Adoption-decision stage

During the adoption-decision stage, as stated by Upadhyay (2020), the organization performs three distinct activities. The initial action is to decide whether to accept or reject the use of BT. Having a favorable attitude toward an innovation does not necessarily result in the decision to adopt it (Rogers, 2003). At this stage, the company must make a careful decision by evaluating the alignment of BT with its strategic objectives, assessing the cost-benefit ratio of implementation, and comparing the substantial advantages of BT with other available solutions.

The second activity at this step involves assessing the proposed blockchain based on technical, financial, and strategic criteria. An in-depth examination of these three aspects enables the organization to ascertain the viability and potential impacts of using BT. In the third activity to be performed at this stage, the organization will determine the required resources for acquiring and implementing BT. These resources encompass both financial and human investments required to acquire, implement, and maintain BT operating in the organization. Prioritizing the assessment of return on investment (ROI) is essential since BT should be self-sustaining; otherwise, its implementation would not be justified (Mundra, 2018; Prasad *et al.*, 2018).

2.1.3 Implementation stage

At this stage, the organization transitions from making creative decisions and generating ideas to implementing technology, which involves a more bureaucratic approach (Rogers, 2003). The initial activity in this stage involves preparing for the acquisition of BT. This involves establishing contractual agreements with BT provider businesses (if the solution development is outsourced) or finalizing revisions with the internal solution development team (if the organization chooses to develop the solution in-house). At this point, it is crucial to provide adequate integration with the organization's current systems (interoperability), and precisely establish implementation prerequisites, such as specific features, security and performance requirements.

Subsequently, the organization must make necessary arrangements to accommodate the utilization of BT. For the successful integration of their daily activities and functions with BT, employees, and stakeholders must undergo preparation to acquire the necessary skills to efficiently utilize the new technology (Ghode *et al.*, 2020b; Queiroz and Fosso Wamba, 2019). To effectively utilize BT, the organization has to obtain relevant expertise and establish the necessary infrastructure to accurately assess its existing resources and determine their intended use. The available alternatives include providing training to the current team, as well as employing new personnel, or outsourcing the required tasks (Wong *et al.*, 2020b).

During the third activity of this stage, the organization is required to conduct a test before the complete implementation of the technology. This test is often conducted through either a pilot test or a Proof of Concept (PoC) (Ghadge *et al.*, 2023). This activity is conducted to validate the functionality of the technology in a smaller and controlled environment and to assess performance, user-friendliness, interoperability and key difficulties encountered by employees (Ghadge *et al.*, 2023).

After evaluating BT's performance during the testing phase, if the results are satisfactory, the organization must decide whether it will officially accept BT. In this manner, the organization can formalize BT's approval for its full use in all of its operations. Finally, the company will be able to continue utilizing the technology in its operations. It is crucial to emphasize that, as previously mentioned, the decision to discontinue the use of the innovation may be made due to disenchantment or replacement in conflicting cases during the confirmation stage (Rogers, 2003).

Consequently, we highlight the significance of meticulously examining the barriers and strategies for their mitigation, while also considering the criticality of each of these phases and activities, which may serve as indicators of the abandonment of the innovation concept.

3. Research method

Our research methodology employs a comprehensive and integrated approach, combining systematic literature review (SLR), content analysis and expert interviews (Figure 2). This methodological triangulation ensures a deep understanding of the subject. We detail this approach in the next sections.

3.1 Systematic literature review

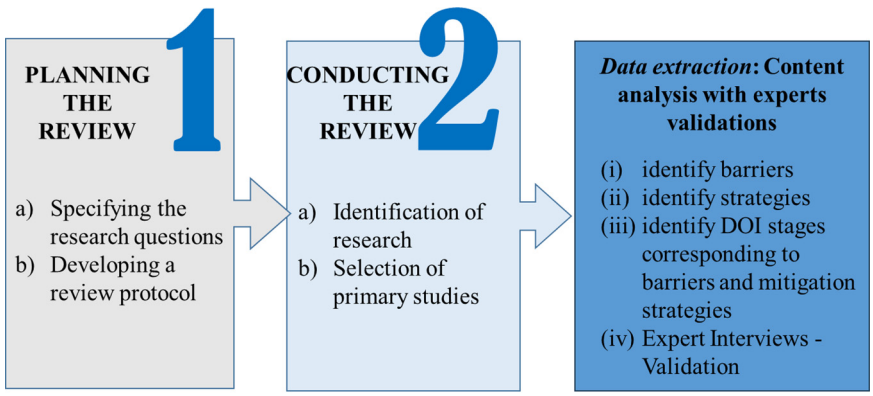
This SLR process is based on the process defined in Tranfield *et al.* (2003) and Kitchenham and Charters (2007). The review protocol was developed based on the PRISMA (Preferred Reporting Items for Systematic Reviews and Meta-Analyses) recommendations (Moher *et al.*, 2015).

3.1.1 Planning the review

In the *planning* step, we first defined the articles' research questions (RQs), which will drive the entire systematic review methodology (Kitchenham and Charters, 2007). Three RQs were proposed in the introduction section.

The review protocol, essential for systematic reviews, ensures thorough planning and documentation, reducing researcher bias and aligning with PRISMA recommendations (Kitchenham and Charters, 2007; Moher *et al.*, 2015). It includes research topics, search strategy and inclusion/exclusion criteria. Our review protocol specified the scientific databases, aligning with Webster and Watson's suggestion that leading journals are key information sources (Webster and Watson, 2002). Hence, Web of Science and Scopus were selected for their extensive journal coverage (Gavel and Iselid, 2008; Singh *et al.*, 2021), while Engineering Village, the central platform for Compendex, was chosen for its comprehensive engineering literature (Dressel, 2017). The three databases mentioned are commonly utilized in contemporary literature reviews within the field of Operations Management (OM) and SCM, as the studies conducted by

Figure 2 Research methodology: SLR, content analysis and expert validation



Source: Authors own work

Corallo *et al.* (2024), Saihi *et al.* (2023), Vazquez Melendez *et al.* (2024) and Swink *et al.* (2024).

We conducted pilot searches to define search terms, observing that articles addressing barriers may not explicitly mention terms like "barrier," "challenge" or "obstacle." Instead, they may use various terms. To avoid limiting results, we opted for words related to blockchain and the context of our study (OM and SC&L). The sets of keywords and search strings used to search for articles in the databases are presented in Table 1.

The review protocol also established inclusion/exclusion criteria. Articles meeting the following conditions presented in Table 1 were included.

3.1.2 Conducting the review

In the *conducting* step, we performed searches in the databases, using the review protocol, and 2,265 articles were initially obtained. 596 duplicate papers were removed (see Figure 3).

As not all of the remaining 1,699 articles were necessarily related to our RQs, assessing their relevance through an article screening process was necessary. In screening relevant papers, the 1,669 articles were initially analyzed based on title, abstract and keywords. The review protocol returned articles with a purely technical, mathematical and computing science focus, which was clearly out of the scope of our study; thus, we excluded them. In some cases, it was difficult to analyze the article's relevance just by reading the title, abstract, and keywords. In this case, we passed it through to the next stage for further reading, allowing us to retain as many relevant articles as possible. Two reviewers conducted this screening to decrease subjectivity in the choice. In this phase, 1,426 papers were rejected, and 243 papers were retained for a full review. The papers approved in all 4 inclusion criteria mentioned in the review protocol were selected for the next step.

3.2 Content analysis with expert panel validation

In our research, we meticulously employed content analysis (CA) (Krippendorff, 2019), as an essential technique to identify technological barriers to BT adoption, the strategies proposed in the literature to mitigate these barriers, and the stages of BT adoption, as delineated by the Diffusion of Innovation (DOI) theory, where these elements manifest. This comprehensive coding process was conducted using Nvivo software, which allowed us to systematically organize and analyze the content of the selected papers. Through this method, we could distill the vast array of information into manageable codes, facilitating a granular examination of the barriers and strategies within the context of BT adoption stages. This process enhances our analysis's precision and depth.

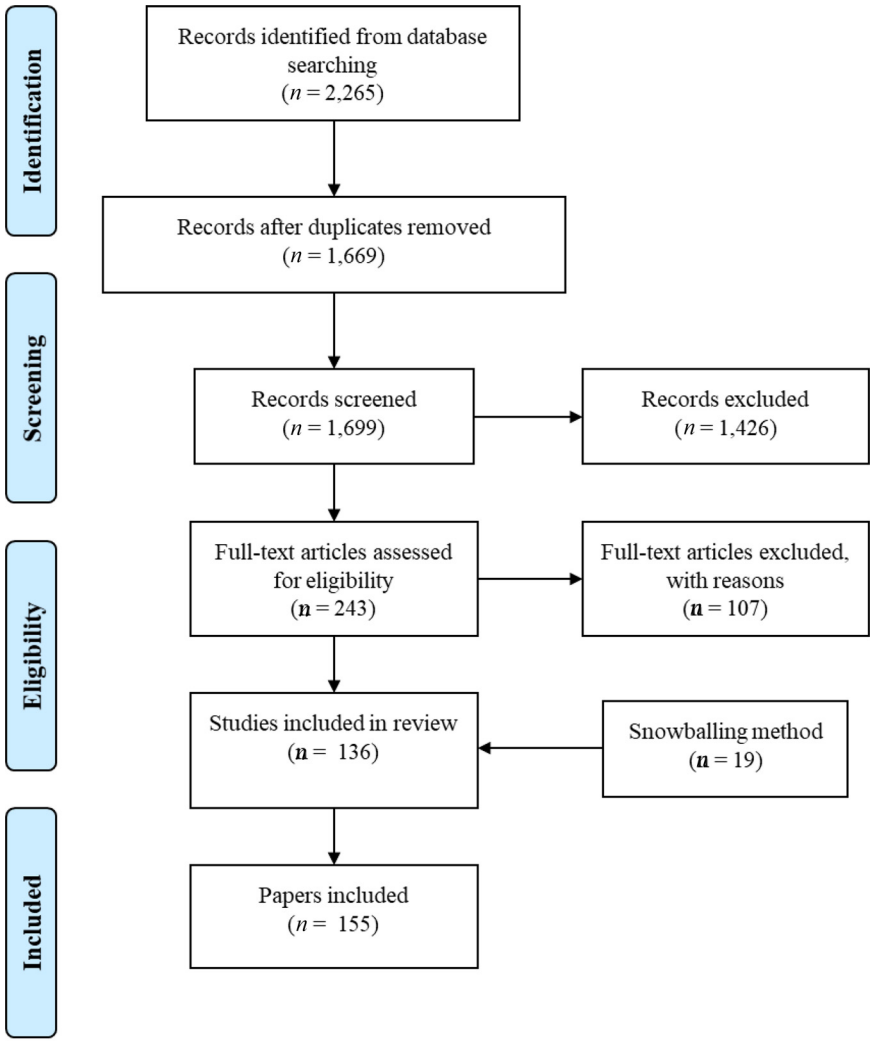
For this coding, we first selected the categories that would be analyzed in each article. To define these categories, we first conduct a keywording step. Other blockchain studies conducted keywording by reading the abstract (Mazlan *et al.*, 2020; Yli-Huumo *et al.*, 2016). As our central research

Table 1 Search terms, data base and include criteria

Keywords	"Blockchain," "BCT," "Distributed Ledger Technology" and "DLT," and another set focusing on the study context, encompassing terms like "supply chain," "value chain," "supply network," "operations," "management," "production," "logistics"
Data base	Web of Science, Scopus, Engineering Village
Search strings	("blockchain" OR "BCT" OR "distributed ledger* technology" OR "DLT") AND ("supply chain" OR "value chain*" OR "supply network*" OR "management" OR "operations" OR "production" OR "logistics")
Include criteria	(1) articles addressing limiting factors to blockchain adoption in SCM and mitigation strategies, (2) articles published in journals, (3) articles written in English, and (4) articles accessible in full

Source: Authors' own work

Figure 3 PRISMA flow diagram



Source: Authors own work

topic is related to technological barriers, the abstract would not always reveal relevant terms. We found that the papers can mention and discuss barriers in other fields of work, even though this is not the final objective of the study. Thus, the keywording was conducted differently, i.e. through an exploratory analysis of 15 articles that mentioned the term barriers or challenges in the title, to generate an initial list of keywords to guide the identification process. These 15 studies were part of the 243 articles selected for the full review. We used the keywords found in this step to form the analyzed categories.

After defining the categories, we conducted a material evaluation step, in which all selected articles were analyzed according to the dimensions established in the category selection. After this full review, we noticed that 107 articles did not address topics related to our research theme. Therefore we excluded them from our sample, and 136 articles were retained, and another 19 were included using the backward snowballing method described by Wohlin (2014). Thus, a total of 155 articles were comprehensively analyzed utilizing CA. This process allowed new keywords to emerge and to be added to the initial categories. The coding cycles were repeated whenever a new relevant keyword appeared.

To ensure the validity and robustness of our findings derived from the coding process, we engaged in a two-tier validation process involving both PhD students and academic experts. Initially, our findings, including the identified barriers, strategies, and their corresponding stages of emergence within the DOI framework, were presented to 16 PhD students in Industrial Engineering, who are involved in the fields of Operations Management (OM) and Supply Chain Management (SCM). This preliminary review by the doctoral candidates served as a pre-refinement step, providing fresh insights and perspectives that contributed significantly to the organization and clarity of our analysis.

Following this initial validation, we further subjected our findings to a rigorous review by four academic experts, as detailed in Table 2. This involved a combination of written

Table 2 Profile of experts

Expert	Position	Academic education	Research area of expertise	Experience with blockchain
1	Researcher/Professor	PhD in industrial engineering	Operations management and supply chain management	5
2	Researcher/Professor	PhD in industrial engineering	Operations management and supply chain management	5
3	Researcher/Professor	PhD in industrial engineering	Operations management and supply chain management	5
4	Researcher/Professor	PhD in engineering	Operations management and supply chain management	6

Source: Authors' own work

assessments and in-depth discussions during scheduled meetings, allowing for a comprehensive critique of our methodology and outcomes. Through individual interviews with each expert, we meticulously reviewed the initial categorization of barriers, strategies and their stages of emergence based on the DOI theory. This expert review process validated the accuracy of our coding and offered opportunities for reassignment and refinement based on expert consensus, revealing intricate interconnections among the adoption stages. This dual-layer validation process, encompassing both PhD students and academic experts, was instrumental in reinforcing the credibility and reliability of our research findings. It ensured that our systematic categorization of barriers, mitigation strategies and their DOI theory-based staging was grounded in the literature and resonated with current academic thought and practical insights.

4. Results and discussion

Previous research has pointed out that technological barriers are only one type of barrier, together with other challenges, such as organizational and user-related barriers and operational barriers, which are also critical for blockchain to work in the SC (Gonczol *et al.*, 2020; Wang *et al.*, 2019a). The function of humans in the adoption of BT is closely correlated with organizational and user-related barriers.

While human resistance may not be a technical issue, it should be given equal importance to the technology barriers discussed in this paper. People are crucial in overcoming technological barriers, as the decision-making process is carried out by individuals who represent the organization. Lokuge *et al.* (2019) assert that the implementation of innovations is influenced not only by the modernity of technology but also by the involvement of IT decision-makers and the existing organizational culture.

4.1 The technological barriers in each step of blockchain technology adoption process

BT faces several technical challenges (Zheng *et al.*, 2018). No company can be fully decentralized without facing security, performance and scalability issues (Iftekhar *et al.*, 2020). In this study, barriers are considered issues directly related to the use of BT that can prevent its commercial use (Kouhizadeh *et al.*, 2021).

Regarding RQ_1, 16 barriers were identified. Each barrier received a code "TB" (technological barrier), ranging from TB_1 to TB_{16}. Table 3 presents such barriers, their definitions, the number of papers dealing with these barriers and the references.

Although all the barriers mentioned in Table 3 are considered technological issues, the majority of them are related to humans. These barriers can generate a certain reluctance to adopt BT, as they have profound implications on human perceptions. Overcoming these technological barriers entails not only technical solutions but also approaches that seek acceptance, trust and human adaptation to the technology.

The *fork problems* barrier can cause resistance in human acceptability as a result of uncertainties regarding the stability and security of the system. The risk of incompatibility with other systems is a result of the *lack of interoperability and standardization* barrier, which generates uncertainty. In sectors where confidentiality is essential, decision-makers may exhibit resistance to *privacy issues*. When irreversible human error is a concern, *immutability* can be perceived as an impediment. The *black-box effect* can lead to reluctance among users due to a lack of comprehension of the fundamentals of BT's operation, which can incite skepticism.

Hacks and vulnerabilities are significant *security concerns* that have the potential to erode the trust of users and organizations in BT. The *lack of tools for BT implementation* can serve as a deterrent to developers and organizations. Due to *usability problems*, users may find it challenging to interact with blockchain applications in a manner that is both intuitive and efficient.

Specialized skills and additional resources may be necessary due to the *increased complexity of IT handling*. The upskilling and technical training or hiring of staff is certainly a huge and important step in any implementation of such importance. The deployment of BT means that an organization needs profound modification of its processes, information systems, and routines, and hence represents a major effort in change management. The blockchain protocol level is distinguished by iterative exchanges between human agents and the blockchain protocol, resulting in a strong interconnection between the technical and human aspects that cannot be separated (Rossi *et al.*, 2019).

The majority of these barriers are associated with the individual who is crucial in the acceptance and decision to adopt BT. This relationship refers to characteristics such as human resistance, skepticism, reluctance, and lack of trust when confronted with barriers that challenge the organization's or the individual's capacity to surmount them. Hence, it is imperative to acknowledge that the technical and human dimensions are interconnected and indivisible. To ensure that blockchain has a tangible effect in the real world, it is crucial to comprehend the social and behavioral factors and difficulties associated with deploying blockchain systems, as explored in the works of Schmidt and Wagner (2019) and Köhler and Pizzol (2020).

Our examination of technological barriers prompted a discussion on the context of BT adoption, aiming to correlate

Table 3 Technological barriers for BT adoption in SCM

Barrier	Definition	References	Papers quantity
Fork problems (TB1)	Forks can occur due to protocol or control code changes, requiring consensus among all nodes (Alam Khan *et al.*, 2020). Failure to reach consensus leads to a bifurcation, where upgraded and nonupgraded nodes coexist, resulting in parallel blocks (Pirrong, 2019)	Alam Khan *et al.* (2020), Andoni *et al.* (2019), Babich and Hilary (2020), Demir *et al.* (2020), Frizzo-Barker *et al.* (2020), Mendling *et al.* (2018), Pirrong (2019), Upadhyay (2020), Yli-Huumo *et al.* (2016), Ziolkowski *et al.* (2020)	11
Lack of interoperability and standardization (TB2)	Challenges in sharing information across diverse blockchain systems due to the absence of universal standards, leading to fragmentation and complexity (Behnke and Janssen, 2020; Chang *et al.*, 2020; Iftekhar *et al.*, 2020; Yadav *et al.*, 2020)	Ahl *et al.* (2020), Al-Jaroodi and Mohamed (2019), Allen *et al.* (2019), Andoni *et al.* (2019), Azaria *et al.* (2016), Babich and Hilary (2020), Bai and Sarkis (2020), Behnke and Janssen (2020), Bodkhe *et al.* (2020), Casino *et al.* (2021), Chang *et al.* (2020), Chen *et al.* (2020), Demir *et al.* (2020), Drljevic *et al.* (2020), Feng *et al.* (2020), Fosso *et al.* (2020 b), Ghode *et al.* (2020b), Gonczol *et al.* (2020), Hackius and Petersen (2020), van Hoek (2019a (2019 b), Hughes *et al.* (2019), Iftekhar *et al.* (2020), Justinia (2019), Kouhizadeh *et al.* (2020), Kurpjuweit *et al.* (2021), Lacity (2018), Liu *et al.* (2019), Mattke *et al.* (2019), Mazzei *et al.* (2020), McGhin *et al.* (2019), Meng *et al.* (2018), Morkunas *et al.* (2019), Olsen and Tomlin (2020), Padmavathi and Rajagopalan (2019), Pólvora *et al.* (2020), Pournader *et al.* (2020), Prasad *et al.* (2018), SA and Umamakeswari (2018), Sukheja *et al.* (2019), Tezel *et al.* (2020), Vilkov and Tian (2019), Wang *et al.* (2019b), Wong *et al.* (2020a), Yadav *et al.* (2020), Yang (2019), Zamani and Giaglis (2018)	47
Privacy problems (TB3)	Reluctance to share confidential information and risks of de-anonymization and transaction pattern exposure necessitates robust protection (Feng *et al.*, 2019; Galvez *et al.*, 2018; Wang *et al.*, 2019b)	Al-Jaroodi and Mohamed (2019), Andoni *et al.* (2019), Babich and Hilary (2020), Behnke and Janssen (2020), Biswas and Gupta (2019), Chen *et al.* (2020), Clohessy and Acton (2019), Drljevic *et al.* (2020), Feng *et al.* (2020 (2019), Fosso *et al.* (2020 b), Fu and Zhu (2019), Ghode *et al.* (2020b), Gonczol *et al.* (2020), van Hoek (2019a (2019 b), Huang *et al.* (2019), Hughes *et al.* (2019), Hussien *et al.* (2019), Jiang *et al.* (2019), Justinia (2019), Kamble *et al.* (2019), Kamilaris *et al.* (2019), Khezr *et al.* (2019), Klöckner *et al.* (2020), Kodym *et al.* (2020), Kosba *et al.* (2016), Kumar *et al.* (2020), Kurpjuweit *et al.* (2021), Li *et al.* (2021), Liu *et al.* (2020), Mackey *et al.* (2019), Mattke *et al.* (2019), McGhin *et al.* (2019), Meng *et al.* (2018), Meyer *et al.* (2019), Montecchi *et al.* (2019), Padmavathi and Rajagopalan (2019), Pólvora *et al.* (2020), Prasad *et al.* (2018), Rane and Narvel (2019), Roehrs *et al.* (2017), Salman *et al.* (2019), Schmidt and Wagner (2019), Sheel and Nath (2020), Srivastava *et al.* (2019), Sternberg *et al.* (2021), Sukheja *et al.* (2019), Tandon *et al.* (2020), Tezel *et al.* (2020), Tiscini *et al.* (2020), Upadhyay (2020), Vilkov and Tian (2019), Wang *et al.* (2019b), Wong *et al.* (2020a (2020 b), Xu *et al.* (2017 (2020), Yadav *et al.* (2020), Yoo and Won (2018), Zheng *et al.* (2019 (2018)	62
Lack of reliability in input data (TB4)	Challenges in ensuring the reliability of data entered into blockchain, as BT can verify data accuracy but not its quality (Howson, 2020; Lu *et al.*, 2019; Schmidt and Wagner, 2019)	Allen *et al.* (2019), Babich and Hilary (2020), Boukis (2019), Chen *et al.* (2020), Galvez *et al.* (2018), van Hoek (2019a (2019 b), Howson (2020), Kamble *et al.* (2020), Khezr *et al.* (2019), Lin (2019), Lu *et al.* (2019), Olsen and Tomlin (2020), Rogerson and Parry (2020), Schmidt and Wagner (2019), Tezel *et al.* (2020), Wang *et al.* (2019b), Wong *et al.* (2020b), Ziolkowski *et al.* (2020)	19
Immutability (TB5)	Recorded data cannot be altered, which can limit adaptability and pose challenges in cases of disputed transactions or data entry errors (Ghode *et al.*, 2020a; Hofmann *et al.*, 2017; Lo *et al.*, 2018)	Behnke and Janssen (2020), Biswas and Gupta (2019), Chen *et al.* (2020), Clohessy and Acton (2019), Farooque *et al.* (2020), Ghode *et al.* (2020a (2020 b), Hackius and Petersen (2020), Hughes *et al.* (2019), Kouhizadeh *et al.* (2021), Mariappan	19

(***continued***)

Table 3 Continued.

Barrier	Definition	References	Papers quantity
		(2019), Padmavathi and Rajagopalan (2019), Pólvora *et al.* (2020), Pournader *et al.* (2020), Saberi *et al.* (2019b), Tiscini *et al.* (2020), Upadhyay (2020), Yoo and Won (2018), Ziolkowski *et al.* (2020)	
Black-box effect (TB6)	Users must trust the process's integrity, even without technical knowledge. Trust extends to specific implementations involving third parties not part of the SC (Babich and Hilary, 2020)	Babich and Hilary (2019 (2020), Fosso *et al.* (2020 b), Sternberg *et al.* (2021)	4
Security problems (TB7)	Vulnerabilities to hacker attacks, 51% attack, key stolen/lost, inaccurate information spread and access to confidential data, exacerbated by blockchain's operation across multiple platforms (Al-Jaroodi and Mohamed, 2019; Kouhizadeh *et al.*, 2021)	Ahl *et al.* (2020), Al-Jaroodi and Mohamed (2019), Alam Khan *et al.* (2020), Behnke and Janssen (2020), Biswas and Gupta (2019), Boukis (2019), Clohessy and Acton (2019), Demir *et al.* (2020), Drljevic *et al.* (2020), Farooque *et al.* (2020), Frizzo-Barker *et al.* (2020), Fu and Zhu (2019), van Hoek (2019a (2019 b), Huang *et al.* (2019), Hughes *et al.* (2019), Hui *et al.* (2019), Hussien *et al.* (2019), Kamble *et al.* (2020), Kamilaris *et al.* (2019), Khezr *et al.* (2019), Kim (2020), Kodym *et al.* (2020), Kouhizadeh *et al.* (2020 (2021), Kuo *et al.* (2017), Kurpjuweit *et al.* (2021), Lee *et al.* (2019), Lin (2019), Lin *et al.* (2017), Lu *et al.* (2019), Mackey *et al.* (2019), Makridakis and Christodoulou (2019), Matsuo (2017), Mazzei *et al.* (2020), McGhin *et al.* (2019), Mendling *et al.* (2018), Padmavathi and Rajagopalan (2019), Pólvora *et al.* (2020), Prasad *et al.* (2018), Prashar *et al.* (2020), Roehrs *et al.* (2017), SA and Umamakeswari (2018), Saberi *et al.* (2019a), Saberi *et al.* (2019b), Srivastava *et al.* (2019), Sukheja *et al.* (2019), Tandon *et al.* (2020), Tiscini *et al.* (2020), Upadhyay (2020), Vilkov and Tian (2019), Viriyasitavat *et al.* (2019), Wang *et al.* (2017), Wang *et al.* (2019b), White (2017), Wong *et al.* (2020a), Yang *et al.* (2020), Yli-Huumo *et al.* (2016), Zhang *et al.* (2018), Zheng *et al.* (2019)	60
High latency (TB8)	Higher response time per transaction in blockchain compared to traditional systems, influenced by factors like network delays and transaction rates (Kuzlu *et al.*, 2019; Yasaweerasinghelage *et al.*, 2017; Yli-Huumo *et al.*, 2016)	Ahl *et al.* (2020), Alam Khan *et al.* (2020), Alrubei *et al.* (2020), Andoni *et al.* (2019), Bodkhe *et al.* (2020), Croman *et al.* (2016), Feng *et al.* (2020), Fosso *et al.* (2020 b), Frizzo-Barker *et al.* (2020), Gonczol *et al.* (2020), Hughes *et al.* (2019), Hussien *et al.* (2019), Jamil *et al.* (2019), Lin *et al.* (2017), Mao *et al.* (2019), Mendling *et al.* (2018), Meng *et al.* (2018), Meyer *et al.* (2019), Morkunas *et al.* (2019), Padmavathi and Rajagopalan (2019), Panarello *et al.* (2018), Pournader *et al.* (2020), Prasad *et al.* (2018), Prashar *et al.* (2020), Salman *et al.* (2019), Seok *et al.* (2019), Tandon *et al.* (2020), Tiscini *et al.* (2020), Wang *et al.* (2017), Xu *et al.* (2017), Yli-Huumo *et al.* (2016), Younan *et al.* (2020), Zhao *et al.* (2019), Zheng *et al.* (2019)	36
Scalability problems (TB9)	Limited storage capacity and transaction processing capabilities strain blockchain's capacity as transaction volume and user data increase (Azzi *et al.*, 2019; Biswas and Gupta, 2019; Kouhizadeh *et al.*, 2021; Kuzlu *et al.*, 2019)	Al-Jaroodi and Mohamed (2019), Alrubei *et al.* (2020), Andoni *et al.* (2019), Behnke and Janssen (2020), Biswas and Gupta (2019), Bodkhe *et al.* (2020), Casino *et al.* (2021), Chang *et al.* (2020), Chen *et al.* (2020), Croman *et al.* (2016), Demir *et al.* (2020), Drljevic *et al.* (2020), Fosso *et al.* (2020 b), Frizzo-Barker *et al.* (2020), Hald and Kinra (2019), Hang *et al.* (2019), van Hoek (2019b (2019a), Hughes *et al.* (2019), Hussien *et al.* (2019), Jawaji *et al.* (2020), Jiang *et al.* (2019), Justinia (2019), Kim (2020), Kouhizadeh *et al.* (2019), Kuo *et al.* (2017), Kurpjuweit *et al.* (2021), Lacity (2018), Lin (2019), Lin *et al.* (2017), Makridakis and Christodoulou (2019), Mao *et al.* (2019), Mazlan *et al.* (2020), Mazzei *et al.* (2020),	68

(***continued***)

Table 3 Continued.

Barrier	Definition	References	Papers quantity
		McGhin *et al.* (2019), Meyer *et al.* (2019), Min (2019), Murari and Ravishankar (2020), Otte *et al.* (2020), Padmavathi and Rajagopalan (2019), Pally and Reddy (2019), Panarello *et al.* (2018), Prasad *et al.* (2018), Rane and Narvel (2019), SA and Umamakeswari (2018), Salman *et al.* (2019), Sanka and Cheung (2019), Seok *et al.* (2019), Shojaei *et al.* (2020), Sukheja *et al.* (2019), Tan *et al.* (2020), Tandon *et al.* (2020), Tezel *et al.* (2020), Tiscini *et al.* (2020), Upadhyay (2020), Vatankhah Barenji *et al.* (2020), Viriyasitavat *et al.* (2019), Xu *et al.* (2017), Yadav *et al.* (2020), Yang (2019), Yang *et al.* (2020), Yli-Huumo *et al.* (2016), Younan *et al.* (2020), Yu *et al.* (2019), Zhang *et al.* (2018), Zhao *et al.* (2019), Zheng *et al.* (2019 (2018)	
Need for high computational speed/ processing power (TB10)	PoW consensus algorithms require substantial computational resources, exceeding the capabilities of many IoT devices and raising sustainability concerns (Seok *et al.*, 2019)	Bodkhe *et al.* (2020), Fosso *et al.* (2020 b), Grigoreva *et al.* (2019), van Hoek (2019b (2019a), Hussien *et al.* (2019), Kamble *et al.* (2020), Mundra (2018), Salman *et al.* (2019), Seok *et al.* (2019), Shojaei *et al.* (2020), White (2017), Yadav *et al.* (2020)	13
Lack of tools for BT implementation (TB11)	Absence of standardized solutions for product tagging complicates the integration of physical and digital worlds (Kouhizadeh *et al.*, 2021; Kurpjuweit *et al.*, 2021)	Farooque *et al.* (2020), Gonczol *et al.* (2020), Kouhizadeh *et al.* (2021), Kurpjuweit *et al.* (2021), Saberi *et al.* (2019a)	5
Usability problems (TB12)	Complexity of blockchain's underlying structure and cryptography principles poses challenges for average users, hindering technology adoption (Wong *et al.*, 2020a; Zamani and Giaglis, 2018)	Alam Khan *et al.* (2020), Frizzo-Barker *et al.* (2020), Hackius and Petersen (2020), Mendling *et al.* (2018), Wang *et al.* (2019b), Wong *et al.* (2020a), Yli-Huumo *et al.* (2016), Zamani and Giaglis (2018)	8
Underperforming functionality (TB13)	Blockchain networks often exhibit inefficiencies by design, with high energy and bandwidth requirements, leading to lower performance compared to centralized databases (Babich and Hilary, 2020)	Babich and Hilary (2020), Rane and Narvel (2019)	2
Complexity in system design (TB14)	The architecture, structure and implementation of blockchain-based systems are complex, which hinders their understanding and implementation (Lakshmi and Sricharan, 2019)	Bodkhe *et al.* (2020), Clohessy and Acton (2019), Fosso Wamba *et al.* (2020b), Hackius and Petersen (2020), Lakshmi and Sricharan (2019), Meng *et al.* (2018), Pereira *et al.* (2019), Perez *et al.* (2020), Rane and Narvel (2019), Upadhyay (2020), Vilkov and Tian (2019), Yadav *et al.* (2020), Zheng *et al.* (2019)	13
Increased IT handling complexity (TB15)	The growing complexity of BT and the need for new IT routines to manage additional interfaces contribute to increased IT handling complexity, necessitating a workforce with enhanced technical expertise and additional resources dedicated to managing and sustaining the current infrastructure (Kamilaris *et al.*, 2019; Sternberg *et al.*, 2021)	van Hoek (2019a (2019 b), Huang *et al.* (2019), Kamilaris *et al.* (2019), Sternberg *et al.* (2021), Wang *et al.* (2019b)	6
Smart contract problems (TB16)	Operational challenges in smart contracts, including limitations in execution, integration within existing systems, inflexibility and errors in coding smart contracts (Fitzhan *et al.*, 2020; Kumar *et al.*, 2020; Tezel *et al.*, 2020)	Cole *et al.* (2019), Feng *et al.* (2020), Fitzhan *et al.* (2020), Hald and Kinra (2019), Kumar *et al.* (2020), Tezel *et al.* (2020), Younan *et al.* (2020)	7

Source: Authors' own work

these barriers with the innovation adoption process. Additional research underscores the necessity for deeper exploration into organizational dynamics and decision-making regarding BT adoption (Frizzo-Barker *et al.*, 2020; Upadhyay, 2020).

Adopting a new idea is challenging, even with obvious advantages (Rogers, 2003). Several studies proposed investigating the organizational innovation adoption process (Frambach and Schillewaert, 2002; Hameed *et al.*, 2012; Jöhnk *et al.*, 2021). Wamba and Queiroz (2020) comment on the need for more efforts to understand the stages of the BT adoption process.

In our research, we identify a relationship between the 16 identified barriers and their emergence across various stages of the BT adoption process (Figure 4). This mapping highlights potential points where organizations might encounter obstacles or consider rejecting BT. By presenting this theoretical model, we aim to equip organizations with a comprehensive understanding of how BT operates. This understanding will enable them to anticipate and effectively navigate the interplay between these barriers and the adoption process.

During the *initiation stage*, companies acknowledge the necessity of blockchain, gain knowledge, form an attitude, and consider adopting BT (Upadhyay, 2020). This stage encompasses the *black-box effect* and *underperforming functionality* barriers. The *black-box effect* in BT allows for easy observation of system inputs and outputs, yet decision-makers may hesitate to adopt it due to challenges in comprehending and visualizing the BT operating process. This theoretical analysis was done (and described in Table 4) for each technological barrier and adoption stage, which resulted in the allocation of barriers presented in our framework.

At the *adoption-decision stage*, companies decide to accept/reject the BT idea, evaluate the technical factors and the financial and strategic perspective, and consider allocating resources for its acquisition and implementation (Upadhyay, 2020). This stage includes the *black-box effect* (also present in the initiation stage), *lack of interoperability and standardization, lack of tools for BT implementation* and *complexity in system design* barriers.

In the *implementation stage*, organizations prepare for the acquisition and use of BT, perform a trial for confirmation and accept and continue its current use (Upadhyay, 2020). This stage includes *lack of interoperability and standardization, lack of tools for BT implementation,* and *complexity in system design* barriers (also present in the adoption-decision stage), *fork problems, privacy problems, lack of reliability in input data, immutability, security problems, high latency, scalability problems, need for high computational speed/processing power, usability problems, increased IT handling complexity* and *smart contract problems*.

4.2 Framework for overcoming technological barriers in blockchain adoption: a phased approach

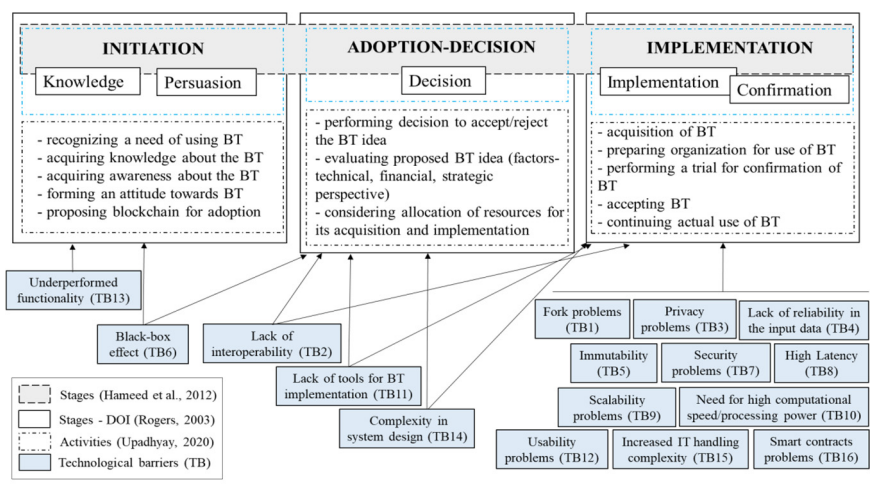
Our study systematically investigates the technological barriers to BT adoption within SCM and outlines strategies to overcome these hurdles. This dual focus allows us to contribute to the existing body of knowledge and offer actionable insights for practitioners in the field. Figure 5 presents a framework that summarizes all the strategies that need to be performed in each phase of the DOI model. Overall, the proposed framework underlines a strategic and phased approach to overcoming the technological barriers to BT adoption, suggesting that each stage of the DOI theory and Hameed *et al.* (2012) presents distinct challenges that can be addressed with targeted strategies to ensure successful integration of BT within SCs.

Our framework presents a comprehensive overview of the mitigation strategies for barriers to the adoption of BT across three key stages – initiation, adoption decision and implementation – guided by the DOI theory.

4.2.1 Initiation phase: laying the foundation for blockchain innovation

The challenge of the *black-box effect* (TB_6) underscores a significant barrier in the blockchain domain where the complexity and opacity of the technology can hinder its understanding and acceptance. While specific strategies to address this barrier directly were not identified in the literature, overcoming it is crucial for enhancing the entire SC. Addressing the *black-box effect* involves demystifying the

Figure 4 Barriers and BT adoption process



Source: Authors own work

Table 4 Allocation of barriers in BT adoption stages

Barriers	Initiation stage	Adoption-decision stage	Implementation stage
Fork problems (TB1)	–	–	Updates to the protocols will be required once the technology is already in use
Lack of interoperability and standardization (TB2)	–	At this stage, the necessary resources for the implementation of BT are analyzed, and the nonpossibility of communication (interoperability) for the transition between centralized systems and BT can interfere with the decision to adopt the technology	The lack of interoperability will interfere with the process of implementing and using BT, hindering cooperation with partners in the supply chain
Privacy problems (TB3)	–	–	There may be problems and hesitation in sharing information while using BT
Lack of reliability in input data (TB4)	–	–	The lack of reliability in the input data is a reality that will be faced during the use of BT, after its implementation
Immutability (TB5)	–	–	The immutability of data can be a problem while BT is in use, due to the lack of flexibility to correct errors and to delete information
Black-box effect (TB6)	Decision-makers may hesitate to proceed with the adoption due to the difficulty of understanding and visualizing the BT operating process	Even if in the initiation stage the decision-maker starts to trust the integrity of a decentralized system, the lack of understanding of its technical fundamentals may get in the way of drawing up a plan for allocating the resources needed for the implementation of BT, at the adoption-decision stage	–
Security problems (TB7)	–	–	Security issues such as vulnerability to hacker attacks will be faced when BT is already in use, after its implementation
High latency (TB8)	–	–	Latency issues will be faced while BT is in use, i.e. after its implementation
Scalability problems (TB9)	–	–	Scalability issues will also be faced while BT is in use, i.e. after its implementation
Need for high computational speed/processing power (TB10)	–	–	The intensive computational consumption will represent a problem when BT is already implemented, resulting in high sustainability costs
Lack of tools for BT implementation (TB11)	–	At this stage, the necessary infrastructure for the adoption of BT is analyzed; the lack of tools for the implementation can be an impediment to the adoption decision	The lack of tools will compromise the effective BT implementation process

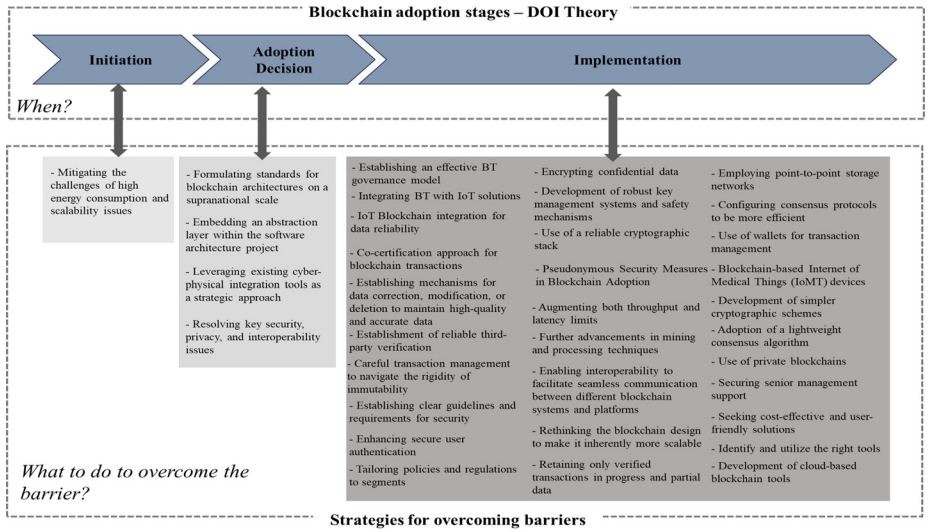
(***continued***)

Table 4 Continued.

Barriers	Initiation stage	Adoption-decision stage	Implementation stage
Usability problems (TB12)	–	–	Difficulties related to BT usability will affect organizations after implementing the technology
Underperforming functionality (TB13)	Decision-makers may notice that the centralized database that the organization uses is more efficient than BT, not recognizing advantages in its substitution, thus declining the decision to adopt	–	–
Complexity in system design (TB14)	–	Such complexity may imply the need for many resources that the company may not have available, which may hinder adoption at this stage	Although the responsibility for developing the system is largely the responsibility of the technology provider company, a joint effort will be required to prepare the organization for testing and using BT, activities that can be affected by the complexity of developing these systems
Increased IT handling complexity (TB15)	–	–	The use of BT will represent a major change in business models that use centralized systems, inserting new routines and increasing the complexity of dealing with information technology
Smart contract problems (TB16)	–	–	Coding of smart contracts will be required while using BT, to regulate trading between companies, so issues related to this barrier will arise at this stage

Source: Authors' own work

Figure 5 A framework for overcoming technological barriers in blockchain adoption



Source: Authors own work

technology to increase awareness, acceptance and adoption among SC members, as emphasized by Meng *et al.* (2018). By making the workings of BT more transparent and comprehensible, stakeholders are more likely to trust and adopt it, thereby facilitating a broader acceptance across the SC network.

Addressing the issue of *underperforming functionality* (TB_{13}) is crucial for enhancing the efficacy of BT within SCs. A key strategy to overcome this barrier, as outlined by Babich and Hilary (2019), involves *mitigating the challenges of high energy consumption and scalability issues* that are often associated with BT. By tackling these problems, BT systems can become more efficient and sustainable, making them more competitive and appealing than traditional systems. Overcoming the underperforming functionality of BT leads to a significant improvement across the entire SC, notably increasing BT adherence, as highlighted by Babich and Hilary (2019).

During the initiation stage, the focus is on mitigating the high energy consumption and scalability issues inherent in BT. It is crucial to take these actions to overcome the barriers that emerge during the initial phase.

We might consider improving energy efficiency to be a useful kind of persuasion since, in addition to decreasing environmental consequences, it may also reduce operational costs for companies, particularly in areas where cost efficiency is critical to competitiveness. Furthermore, BT's scalability may enhance adoption among stakeholders who handle a large volume of transactions. Scalability is critical for ensuring the successful functioning of BT in applications involving SCs, where a large number of organizations interact and exchange information. Solving barriers like high energy consumption and scalability issues may assist businesses develop a more positive attitude toward BT implementation.

4.2.2 Adoption decision phase: evaluating and committing to blockchain

A holistic strategy is paramount to tackle the significant barrier of *lack of interoperability and standardization* (TB_2) in blockchain implementation. The literature posits a dual approach to surmount this obstacle: initially, by *formulating standards for blockchain architectures on a supranational scale*, as advocated by Andoni *et al.* (2019), Allen *et al.* (2019), Yang (2019), Behnke and Janssen (2020), Feng *et al.* (2020), Pólvora *et al.* (2020), Iftekhar *et al.* (2020) and Chen *et al.* (2020); and subsequently, by *embedding an abstraction layer within the software architecture project*, a method discussed by Mazzei *et al.* (2020). Implementing this strategy necessitates a series of concerted actions.

As Zamani and Giaglis (2018) and Pólvora *et al.* (2020) suggested, establishing consortia and alliances among companies lays the foundational groundwork. This strategy is implemented in blockchain interoperability alliance that was founded in a joint effort by companies such as Aion, ICO, Wanchain and others, to improve communication between these systems (Drljevic *et al.*, 2020). These efforts reiterate the need for collaboration from all stakeholders to achieve full interoperability (Feng *et al.*, 2020).

This effort is augmented by middleware integration to facilitate system integration, as recommended by Prasad *et al.* (2018) and Fosso Wamba *et al.* (2020a). This strategy was put into practice in Origin Trail, which implemented an agnostic middleware that provides interoperability between blockchains (Fosso *et al.*, 2020 b). Middleware can also promote the integration of BT with data management systems used in SC&L, such as ERP, Warehouse Management Systems (WMS), Customer Relationship Management (CRM) and Supplier Relationship Management (SRM) (Pournader *et al.*, 2020). Finlync (a privately held global fintech company) developed an agnostic middleware to enable SAP integration for Ethereum and Hyperledger (Prasad *et al.*, 2018).

The formation of a technical committee, as endorsed by Yang (2019) and Allen *et al.* (2019), bolsters the structural framework for standardization. A technical committee of the International Standardization Organization (ISO/TC307) was established to promote the design of technical blockchain

standards (Allen *et al.*, 2019; Eyers, 2018; ISO, 2017; Yang, 2019).

Further, the utilization of blockchain-based systems tailored to resolve interoperability issues in healthcare, as demonstrated by McGhin *et al.* (2019), along with blockchain network projects designed to foster interoperable systems, as evidenced by Mattke *et al.* (2019), Pólvora *et al.* (2020) and Chang *et al.* (2020), is critical. Examples of this strategy employed in this industry include OmniPHR (Roehrs *et al.*, 2017), Medrec (Azaria *et al.*, 2016), Secure System for Pervasive Social Network (PSN) (Zhang *et al.*, 2016) and Fast Healthcare Resources (FHIRChain) (Zhang *et al.*, 2018). The Drug Supply Chain Security Act (DSCSA) has called for an interoperable electronic system to be developed, from which it will be possible to track drugs automatically in the SC (Mattke *et al.*, 2019).

The Cosmos Network and The Polkadot Network are projects for creating a blockchain network (Chang *et al.*, 2020). These technologies propose validators as bridges between different blockchain environments, such as Oraclize (Chang *et al.*, 2020). Moog is developing VeriPart, a blockchain-based service, and has partnered with Nuco, which is building an interoperable blockchain network called Aion (Lacity, 2018). Moog is also working with a significant supplier so that the platform can also connect to legacy ERP systems (Lacity, 2018).

The incorporation of open-source solutions, as recommended by Hughes *et al.* (2019), also plays a vital role. Organizations are studying the feasibility of approaches using Hyperledger (an *open-source solution* hosted by the Linux Foundation) and Tendermint (a general-purpose blockchain consensus engine able to host arbitrary application states) (Hughes *et al.*, 2019).

Overcoming the *interoperability and standardization* barrier heralds significant enhancements across the SC. It enables data aggregation, permitting synergistic use of information among participants, an improvement emphasized by Babich and Hilary (2020). It also assures compatibility among diverse blockchain systems, as noted by Chang *et al.* (2020) and Ghadge *et al.* (2023), and between blockchains and the prevailing data management systems in SC&L, facilitating real-time information sharing among stakeholders, a crucial development highlighted by Pournader *et al.* (2020).

The challenge of a *lack of tools for BT implementation* (TB_{11}) necessitates innovative solutions to bridge the gap between the physical and digital realms within SCs. The literature suggests *leveraging existing cyber-physical integration tools as a strategic approach* to address this barrier, as discussed by Kurpjuweit *et al.* (2021) and Gonczol *et al.* (2020). The use of paired Radio Frequency Identification (RFID) tags, as recommended by the same authors, offers a tangible solution by providing a seamless link between physical items and their digital counterparts on the blockchain. Additionally, incorporating Nearfield Communication (NFC) technology or barcodes, as suggested by Kurpjuweit *et al.* (2021), further enhances the connection between the physical and digital worlds, facilitating the tracking and authentication of assets throughout the SC.

More information about RFID can be found in Stockman (1948), the combination of RFID and BT in Tian (2016) and Vilkov and Tian (2019) and the combination of BT and NFC in Alzahrani and Bulusu (2018). These technologies enable the creation of natively digital assets, significantly reducing data unreliability and enhancing the integrity of input data, a crucial improvement highlighted by Babich and Hilary (2020).

The challenge of *complexity in system design* (TB_{14}) is a significant barrier to the effective deployment of BT in SCs. Addressing this barrier necessitates a focus on *resolving key security, privacy and interoperability issues*, as identified by Yadav *et al.* (2020). These elements are critical to ensuring that BT systems are robust, user-friendly and capable of seamless integration within diverse SC environments. Overcoming the complexity in BT system design benefits the entire SC and is particularly advantageous for technology providers and developers. By simplifying the design and enhancing the functionality of BT systems, the migration from traditional centralized systems to more decentralized BT-based systems becomes more feasible and attractive, as highlighted by Clohessy and Acton (2019). This transition facilitates a more agile, transparent and secure SCM process.

In concluding our discussion on the adoption decision phase, it's imperative to highlight the emergence of the *black box effect* (TB_6). The presence of this barrier underscores a significant challenge that must be addressed to improve the prospects of successful blockchain implementation. As previously noted, our SLR did not uncover any strategies specifically tailored to directly mitigate this barrier, revealing a notable gap in current research. This absence of targeted solutions for the "black box effect" presents a critical area for future investigation. Addressing this gap is essential for developing comprehensive strategies that enhance transparency and understanding of BTs, thereby facilitating their broader acceptance and integration.

As organizations transition to the Adoption Decision stage, strategies include formulating international standards for blockchain architecture, embedding abstraction layers within software projects, leveraging cyber-physical integration tools, and resolving key issues around security, privacy, and interoperability. At this stage, firms must decide whether or not to embrace BT based on solid information about the alignment of BT with their strategic objectives, evaluating the cost-benefit ratio of implementation, and comparing BT to other available options.

Developing international standards for blockchain architecture assists in reducing the uncertainty and concern of adopting a technology that is still gaining traction, such as BT. Furthermore, established standards can help BT integrate and interoperate with current systems within enterprises, as well as with different blockchain networks throughout the world. Embedding abstraction layers in software projects can help simplify the complexity of BT for both developers and end users. Leveraging cyber-physical integration tools can demonstrate how enterprises may benefit from BT, both in terms of process optimization and as a driver of digital transformation.

Resolving key issues such as security, privacy, and interoperability can reduce companies' hesitation to adopt a technology capable of increasing reliability and trust, particularly in environments such as the SC where there are conflicts of interest. In this context, companies are interested in data protection and regulatory compliance. Addressing all of these challenges at this stage would undoubtedly offer the

organization with more strong justifications for making a positive decision to use BT.

4.2.3 Implementation phase: executing blockchain integration

Addressing the challenge of *fork problems* (TB_1) requires a focused approach. The literature suggests *establishing an effective blockchain governance model* as a pivotal strategy to navigate this issue, as Pirrong (2019) and Ziolkowski *et al.* (2020) highlighted. Implementing such governance can significantly mitigate the risk of blockchain forks, where nodes may diverge into separate chains, thus avoiding bifurcation, a critical improvement underscored by Alam Khan *et al.* (2020) and Babich and Hilary (2020). Moreover, central governance fosters enhanced communication and compatibility across different blockchain versions, whether they are updated or adhere to non-updated protocol versions, as noted by Alam Khan *et al.* (2020) and Babich and Hilary (2020). Resolving fork-related problems facilitates superior integration and communication among SC business entities. This strategic approach directly addresses the *fork problems* (TB_1) and strengthens the overall blockchain infrastructure within the SC, ensuring a more cohesive and efficient operational framework.

Establishing an effective blockchain governance model can be achieved by formulating policies, guidelines, and organizational structures that regulate its usage and management (Zwitter and Hazenberg, 2020). Good governance in liberal democratic institutions typically involves strict adherence to the rule of law. This means that current laws and regulations are applied in a straightforward and unbiased manner, without being influenced by social or political factors (De Filippi *et al.*, 2020).

Confronting *privacy issues* (TB_3), necessitates a nuanced and multifaceted strategy. The literature underscores the importance of *integrating Blockchain Technology (BT) with Internet of Things (IoT) solutions* as an effective approach to mitigating privacy concerns, as suggested by Justinia (2019). To operationalize this strategy, a series of targeted actions are proposed in the literature:

- employing decentralized machine learning to ensure minimal privacy infringement (Liu *et al.*, 2020);
- adopting secure collaborative machine learning frameworks that incorporate differential privacy to safeguard sensitive data (Liu *et al.*, 2020);
- establishing a distributed trusted communication network to enhance secure exchanges (Li *et al.*, 2021);
- developing a distributed and reliable cloud data source architecture that leverages BT and ProvChain for enhanced data integrity and privacy (Yoo and Won, 2018);
- utilizing Federated or Consortium Blockchains to control access and improve privacy (Padmavathi and Rajagopalan, 2019; Zheng *et al.*, 2019);
- implementing key agreement protocols like Elliptic Curve Diffie-Hellman-Merkle (ECDHM) to secure communications (Sukheja *et al.*, 2019);
- introducing decentralized smart contract systems, such as Hawk, that preserve privacy while executing contracts (Fu and Zhu, 2019; Kosba *et al.*, 2016; Zhao *et al.*, 2019);
- creating access control models and data authority management methods to regulate data access and ownership (Huang *et al.*, 2019; Jiang *et al.*, 2019);
- ensuring pseudonymity or employing permissioned ledgers to anonymize transactions (Andoni *et al.*, 2019); and
- exploring solutions, prototypes, projects and schemes specifically tailored for the healthcare sector to address its unique privacy requirements (McGhin *et al.*, 2019).

Efforts and techniques to address *privacy issues* include Stealth Addresses, Pedersen Commitments, Ring Signature, Homomorphic Encryption, Zero-Knowledge-Proof (Hamida *et al.*, 2017; Zhao *et al.*, 2019), encrypted transactions (Kosba *et al.*, 2016; Zhao *et al.*, 2019), double blockchain architecture perspective (Zhao *et al.*, 2019), Centralized and Decentralized mixing (Feng *et al.*, 2019; Zheng *et al.*, 2018), CryptoNote and NIZK (Feng *et al.*, 2019).

Salman *et al.* (2019) mentioned Zyskind's approach, Blockchain-Based Data Sharing (BBDS), FairAccess, Dynamic Access Control for IoT Using FairAccess, Decentralized Runtime Access Monitoring System (DRAMS) and Data Privacy for IoT Data Storage and Sharing. Gonczol *et al.* (2020) highlighted Hyperledger Fabric and Permissioned Systems, the recording of off-chain data hashes, role-based access control and Multi-Chain Architectures.

Jiang *et al.* (2019) proposed an access control model where members need to request access to data records. To protect the confidentiality, Huang *et al.* (2019) designed a data authority management method to regulate access to sensor data. When using pseudonymity or allowed ledgers, access to data can be restricted to authorized parties (Andoni *et al.*, 2019).

The healthcare sector presents solutions to improve, among other problems, privacy, with the literature highlighting: Health Data Gateway (HDG) (Al-Jaroodi and Mohamed, 2019; Justinia, 2019; McGhin *et al.*, 2019); a blockchain-based privacy-preserving data sharing (BPDS), a new data preservation system (DPS), a BHEEM framework, and a novel decentralized attribute-based signatures (ABS) scheme, and MediChainTM (Hussien *et al.*, 2019); MedShare (Hussien *et al.*, 2019; McGhin *et al.*, 2019); MedBlock (Hussien *et al.*, 2019; Padmavathi and Rajagopalan, 2019); Fast Healthcare Resources (FHIRChain) (Zhang *et al.*, 2018); A new blockchain-based data scheme (BBDS) for security and privacy preservation for data transactions (Tandon *et al.*, 2020); OmniPHR (McGhin *et al.*, 2019; Roehrs *et al.*, 2017); MediLedger project (Mattke *et al.*, 2019). Further, the Bloodchain prototype, using Hyperledger Iroha and Ardor systems to maintain patient privacy, was developed by Pólvora *et al.* (2020).

Overcoming the barrier of *privacy concerns* not only enhances user confidentiality and preserves sensitive data but also establishes robust information access control mechanisms (Huang *et al.*, 2019; Panarello *et al.*, 2018). This comprehensive approach to addressing privacy in blockchain implementations instills greater confidence among technology providers, business entities, governments and customers in the security and integrity of information shared on the network.

Addressing the challenge of *lack of reliability in input data* (TB_4) demands a strategic approach that incorporates both technological and procedural solutions. The literature advocates for the *IoT Blockchain integration for data reliability* (Allen *et al.*, 2019; Babich and Hilary, 2020; Brookbanks and

Parry, 2024; Olsen and Tomlin, 2020; Rogerson and Parry, 2020; Ziolkowski *et al.*, 2020). This integration or ensuring that data is natively digital makes information collecting more precise and reliable, which is important, especially in situations where data integrity is critical, such as SC&L, and the healthcare sector, among others. These industries can benefit from increased security, traceability, process automation, and decreased fraud thanks to this integration (Agarwal *et al.*, 2022).

Further enhancing data integrity, the *cocertification approach for blockchain transactions*, i.e. certification of transactions by both buyers and sellers, as recommended by Babich and Hilary (2020), ensures a mutual validation process. Certifying transactions by buyers and sellers involves verifying and documenting transactions in a secure and unchangeable manner on the blockchain. This process fosters transparency and confidence by ensuring that all parties participate in confirming the transactions.

The importance of *establishing mechanisms for data correction, modification or deletion to maintain high-quality and accurate data* is underscored by Ziolkowski *et al.* (2020). Additionally, the *establishment of reliable third-party verification*, supported by Babich and Hilary (2020), Howson (2020) and Ziolkowski *et al.* (2020), introduces an additional layer of data validation. Successfully overcoming the barrier of data unreliability not only improves the overall data quality within the system, as indicated by Lu *et al.* (2019), but also contributes to the creation of an incorruptible system, enhancing trust among SC partners, as noted by Babich and Hilary (2020) and Bai and Sarkis (2020).

The challenge of *immutability* (TB_5) requires a discerning approach to ensure the integrity and flexibility of blockchain systems. Biswas and Gupta (2019) suggest the need for *careful transaction management to navigate the rigidity of immutability*. Addressing this barrier effectively can lead to substantial enhancements across the entire SC. By adopting meticulous transaction practices, the system can maintain the foundational principle of immutability while introducing a degree of flexibility to rectify errors, as highlighted by Pournader *et al.* (2020). This approach also facilitates the removal of inaccurate or privacy-violating data, ensuring compliance with data protection standards, as noted by Babich and Hilary (2020).

Addressing *security problems* (TB_7) in BT necessitates a comprehensive and multifaceted approach. The literature outlines several strategies to fortify security measures, beginning with *establishing clear guidelines and requirements*, as suggested by Al-Jaroodi and Mohamed (2019). *Enhancing secure user authentication* is critical for safeguarding access and interactions within the blockchain, a measure advocated by Kim (2020). *Tailoring policies and regulations to segments* requiring heightened protection further strengthens the security framework, as emphasized by Al-Jaroodi and Mohamed (2019). *Encrypting confidential data* is paramount to ensuring privacy and security, a practice supported by Kuo *et al.* (2017) and Zhang *et al.* (2018). The *development of robust key management systems and safety mechanisms*, as discussed by Hussien *et al.* (2019) and Hughes *et al.* (2019), alongside the *use of a reliable cryptographic stack* as proposed by Mazzei *et al.* (2020), are essential for maintaining secure blockchain environments. Frizzo-Barker *et al.* (2020) highlight the importance of **pseudonymous Security Measures in Blockchain Adoption**, i.e. anonymity and the use of pseudonyms to protect user identities.

To implement these strategies, various actions are proposed, including the adoption of *Private blockchains* (e.g. Multichain and Bankchain) ensure confidentiality, privacy and control (Padmavathi and Rajagopalan, 2019; Yang *et al.*, 2020). *Consortium blockchain* (e.g. Hyperledger), which is restricted to the participation of consortium members, improves privacy and security (Feng *et al.*, 2019; Lu *et al.*, 2019).

Formal analysis and verification methods, as discussed by Matsuo (2017) and Zheng *et al.* (2019), alongside key security wallets and consensus mechanisms like Proof-of-Stake (PoS) and credit-based Proof-of-Work (PoW), as suggested by Alam Khan *et al.* (2020), Lin *et al.* (2017), Lu *et al.* (2019) and Huang *et al.* (2019), further enhance the security infrastructure. The Practical Byzantine Fault Tolerance (PBFT) algorithm, as noted by Lu *et al.* (2019) and Sukheja *et al.* (2019), and the integration of virtual private networks or HIPAA-compatible components, as recommended by Kuo *et al.* (2017) and Zhang *et al.* (2018), also contribute to a more secure system.

Additionally, sector-specific solutions, particularly in healthcare, as explored by McGhin *et al.* (2019), address unique security needs. Overcoming the barrier of *security problems* not only ensures data integrity, as highlighted by Panarello *et al.* (2018) but also significantly reduces vulnerabilities to hacker attacks, as noted by Al-Jaroodi and Mohamed (2019). As a result, participants in the SC can rely on a tamper-proof and secure system, fostering trust and reliability in BT's potential to revolutionize SCM. Healthcare industry solutions include MedBlock, which is excellent in terms of privacy and security (Fan *et al.*, 2018; Padmavathi and Rajagopalan, 2019); a new blockchain-based data scheme (BBDS) for security and privacy preservation (Tandon *et al.*, 2020); Trial and Precision Medicine, which adds integrity and security of access to medical data (McGhin *et al.*, 2019); and Healthcare Data Gateways [a smartphone (IoT) application], which uses the decentralized nature of BT to improve security and data integrity (McGhin *et al.*, 2019).

Tackling the issue of *high latency* (TB8) within blockchain systems demands targeted strategies to enhance speed and efficiency. The literature recommends **augmenting both throughput and latency limits** as fundamental steps to mitigate this barrier, as evidenced by the work of Croman *et al.* (2016) and Alam Khan *et al.* (2020). **Further advancements in mining and processing techniques** are crucial for accelerating blockchain transactions, a notion supported by Salman *et al.* (2019) and Seok *et al.* (2019). To actualize these strategies, the adoption of multi-chain architectures is proposed by Gonczol *et al.* (2020) to distribute the load and enhance processing speeds. In multi-chain architectures, SKYFChain, with its three-tier architecture, applies a two-step block generation approach to overcome the latency issues of current PoW-based blockchains (Gonczol *et al.*, 2020). The Bitcoin-NG protocol, highlighted by Eyal *et al.* (2016) and Zhao *et al.* (2019), presents a notable reduction in latency compared to traditional Bitcoin, offering a viable solution to the latency challenge. Additionally, integrating software solutions like BigchainDB, as discussed by BigchainDB GmbH (2018) and

Zheng *et al.* (2019), combines blockchain's robust features with the efficiency of database systems to improve transaction speeds. Exploring alternative consensus mechanisms to the conventional Proof of Work (PoW), as suggested by Alrubei *et al.* (2020) and Ahl *et al.* (2020), and implementing sharding techniques, as discussed by Lin *et al.* (2017) and Ahl *et al.* (2020), further contribute to reducing latency by partitioning the network and data, allowing parallel transaction processing. Addressing the challenge of high latency leads to a more responsive and efficient blockchain system, significantly improving transaction times and reducing delays, as noted by Kuzlu *et al.* (2019).

Confronting *scalability problems* (TB_9) in BT is essential for its widespread adoption and effective functioning within SCs. The literature suggests several strategies to enhance scalability, starting with *enabling interoperability to facilitate seamless communication between different blockchain systems and platforms*, as advocated by Chang *et al.* (2020). *Rethinking the blockchain design to make it inherently more scalable* is crucial, a concept explored by Croman *et al.* (2016). Streamlining the blockchain by *retaining only verified transactions in progress and partial data* can also significantly enhance scalability, as proposed by Kuo *et al.* (2017) and Zhang *et al.* (2018). *Employing point-to-point storage networks*, as suggested by Gonczol *et al.* (2020), and *configuring consensus protocols to be more efficient*, as noted by Xu *et al.* (2017), are additional strategies to address scalability. The *use of wallets for transaction management*, highlighted by Mendling *et al.* (2018), and *blockchain-based Internet of Medical Things (IoMT) devices*, as discussed by Hussien *et al.* (2019), contribute to the scalability solutions.

To implement these strategies, various actions are proposed: adopting open-source protocols to encourage innovation and flexibility, as noted by Kim (2020); utilizing Directed Acyclic Graphs (DAGs) to enhance throughput and consensus speed, as explored by Jiang *et al.* (2019) and Huang *et al.* (2019); and considering Double Chain architecture and private blockchains to improve system efficiency, as suggested by Leng *et al.* (2018) and Prasad *et al.* (2018), respectively. Blockchain compression methods and the adoption of Federated, Consortium, and Permission-based Blockchains, as recommended by Bruce (2018), Hussien *et al.* (2019), Padmavathi and Rajagopalan (2019), Zheng *et al.* (2019), Behnke and Janssen (2020), also play a vital role in addressing scalability. Sharding, which involves partitioning the blockchain into more manageable segments, stands out as a significant technique to improve scalability, with contributions from Yu *et al.* (2019), Meyer *et al.* (2019) and Vinay Reddy (2019), among others. Network sharding is considered one of the most important techniques to solve the blockchain scalability problem and improve blockchain performance (Huang *et al.*, 2022), which can alleviate the uneven distribution of blockchain transactions to a certain extent.

Sharding allows the system to process transactions in parallel, thus improving scalability (Ahl *et al.*, 2020; Andoni *et al.*, 2019; Meyer *et al.*, 2019; Panarello *et al.*, 2018; Yu *et al.*, 2019). Furthermore, it is possible to improve blockchain network throughput through the *RAFT algorithm*, which allows for more than 10,000 transactions (Alam Khan *et al.*, 2020).

Solana, a public blockchain platform that was introduced in April 2018, is designed to enhance scalability in comparison to other blockchains while maintaining decentralization and security (Pierro and Tonelli, 2022). Polkadot is a blockchain system linking a network of blockchains, each designated as a parachain, that may be tailored to the corporate needs of a given application (Scott *et al.*, 2023). Polkadot offers the security, verifiability and trust of a public blockchain together with the cost benefits, scalability and control of a permissioned or private blockchain (Scott *et al.*, 2023).

Leveraging BigChainDB for handling large data volumes, as proposed by Viriyasitavat *et al.* (2019), Gonczol *et al.* (2020) and Alam Khan *et al.* (2020), and optimizing storage and redesigning blockchain structures, as discussed by Zheng *et al.* (2018) and Mazlan *et al.* (2020), further enhance the scalability of blockchain systems. Successfully overcoming scalability challenges not only improves the storage capacity and transaction processing rate of BT, as highlighted by Azzi *et al.* (2019), Kouhizadeh *et al.* (2021), but also expands the platform's capacity to serve a larger number of participants, as noted by Kuzlu *et al.* (2019).

Solutions for storage optimization and redesigning blockchain to solve the block size problem have mostly been found in the healthcare sector:

- Storage optimization: mini-blockchain, (Bruce, 2018; Mazlan *et al.*, 2020; Zheng *et al.*, 2018), VerSum (Van Den Hooff *et al.*, 2014; Mazlan *et al.*, 2020; Zheng *et al.*, 2018), and Reference Pointer FHIRCain (Mazlan *et al.*, 2020; Zhang *et al.*, 2018).
- Redesigning blockchain: FHIRChain (Mazlan *et al.*, 2020; Zhang *et al.*, 2018), HealthChain (Ahram *et al.*, 2017; Mazlan *et al.*, 2020), DeepLinQ (Chang *et al.*, 2019; Mazlan *et al.*, 2020) and Omni PHR (Mazlan *et al.*, 2020; McGhin *et al.*, 2019; Roehrs *et al.*, 2017) (as blockchain modeling solutions); short-term data sharing (Mazlan *et al.*, 2020; Zhang *et al.*, 2018) and caching systems (Mazlan *et al.*, 2020; Sanka and Cheung, 2019) (as reading mechanism solutions); Chord algorithm (Al-Jaroodi and Mohamed, 2019; Mazlan *et al.*, 2020), tokenization (Mazlan *et al.*, 2020; Zhang *et al.*, 2018), sharding (Team Zilliqa, 2017; Mazlan *et al.* (2020), Practical Byzantine Fault (PBFT) consensus protocol (Team Zilliqa, 2017; Mazlan *et al.* (2020), and TrustChain (Mazlan *et al.*, 2020; Otte *et al.*, 2020) (as write mechanism solutions); and Lightning networks (Mazlan *et al.*, 2020; Poon and Dryja, 2016) (as a bidirectional solution).

Addressing the *need for high computational speed and processing power* (TB_{10}) is crucial for enhancing the efficiency and sustainability of blockchain systems within SCs. The literature offers several strategies to mitigate this challenge. Salman *et al.* (2019) propose the *development of simpler cryptographic schemes* to reduce the computational burden associated with blockchain transactions.

Additionally, the **adoption of a lightweight consensus algorithm**, as suggested by Seok *et al.* (2019), can significantly lower the demand for processing power without compromising security and integrity. Proof of Stake (PoS), Delegated Proof of Stake (DpoS) and Proof of Block and Trade (PoBT) are examples of some of the lightweight consensus algorithms. The advantage of PoS is that it eliminates the necessity for significant computing capacity and increased electricity

consumption, as is the case with PoW (Alam Khan *et al.*, 2020). DpoS is a lightweight consensus algorithm that uses a selected number of elected delegates to validate and confirm transactions, therefore reducing the performance and efficiency loss in IoT blockchain-based systems (Haque *et al.*, 2024). PoBT is a solution that enables the verification of trades and blocks with reduced computational time (Biswas *et al.*, 2020).

The **use of private blockchains**, recommended by Cole *et al.* (2019), presents another viable solution by limiting the network size and, consequently, the computational requirements. Such blockchains can be tailored to the specific needs of the SC entities, offering a balance between security, privacy, and computational efficiency. Effectively overcoming this barrier not only reduces wasted computing resources, including hardware and electricity, as highlighted by Khan *et al.* (2020) but also facilitates resource savings across the SC.

Addressing *usability problems* (TB_{12}) within BT is essential for fostering wider acceptance and utilization across SCs. To mitigate this barrier, **securing senior management support** is paramount, as their endorsement can drive the adoption and integration of BT by aligning organizational strategies and resources, a notion supported by (Wong *et al.*, 2020a). Additionally, **seeking cost-effective and user-friendly solutions** is crucial for lowering the entry barriers to BT usage, as emphasized by Hackius and Petersen (2020). Examples that execute these solutions include the "Obsidian" programming language, as discussed by Coblenz (2017) and Alam Khan *et al.* (2020), which offers a more intuitive approach to smart contract development within blockchain systems. Tools such as BitConeView and Bitlodine, highlighted by Yli-Huumo *et al.* (2016), provide user-friendly interfaces for analyzing and visualizing blockchain data, further simplifying the interaction with BT. Successfully overcoming usability challenges not only boosts BT users' confidence by demystifying technical complexities, as noted by Wong *et al.* (2020a) but also enhances BT adherence and proliferation within the SC sector, as highlighted by Zamani and Giaglis (2018).

The issue of *increased IT handling complexity* (TB_{15}), poses a significant challenge to the broader adoption and implementation of BT within SCs. To address this barrier, it is essential **to identify and utilize the right tools**, a strategy underscored by Rane and Narvel (2019). Additionally, the **development of cloud-based blockchain tools**, as advocated by Clohessy and Acton (2019), offers a promising solution to streamline BT management and operation. As suggested by Kamilaris *et al.* (2019), the integration of blockchain oracles further enhances the functionality and efficiency of BT systems by facilitating the secure and reliable interaction between blockchain networks and external data sources. These advancements simplify the IT complexities associated with managing and operating BT systems. Successfully overcoming the barrier of increased IT handling complexity holds significant implications for the entire SC. By reducing the complexity and technical demands of BT, the technology becomes more accessible to a wider range of organizations, particularly small and medium-sized enterprises (SMEs) that may have limited IT resources. This accessibility, as highlighted by Clohessy and Acton (2019), not only facilitates the democratization of BT but also promotes its wider dissemination across various sectors of the SC.

The challenge posed by *smart contract problems* (TB_{16}) significantly impacts the efficacy and reliability of BT within SCs. Although the literature does not provide specific solutions to this barrier, addressing it is crucial for the seamless operation of smart contracts, which are pivotal for automating transactions and agreements within the blockchain. Overcoming the issues related to smart contracts would lead to considerable improvements across the entire SC. One of the primary benefits would be enhanced negotiations among the various stakeholders within the SC. By reducing the vulnerability to code errors in smart contracts, as highlighted by Feng *et al.* (2020), Kumar *et al.* (2020), and Tezel *et al.* (2020), the integrity and reliability of these digital contracts are significantly improved. This enhancement in reliability fosters trust among parties and streamlines the execution of agreements without the need for intermediaries. Moreover, addressing smart contract problems facilitates the automation of processes, making operations more efficient and reducing the likelihood of conflicts between parties. The automation enabled by reliable smart contracts can transform SCM by making it more transparent, efficient, and secure.

In wrapping up our analysis of the implementation phase, it's crucial to acknowledge that barriers to *lack of interoperability and standardization* (TB_2), *lack of tools for BT implementation* (TB_{11}), and *complexity in system design* (TB_{14}) also manifest during this stage. Given their presence, it becomes imperative to apply the strategies identified in earlier phases that are designed to counteract these specific barriers. The continuity in addressing the *lack of interoperability and standardization* (TB_2), *lack of tools for BT implementation* (TB_{11}) and *complexity in system design* (TB_{14}) throughout the implementation phase underscores the importance of leveraging previously outlined mitigation strategies. Employing these strategies within this phase is essential for navigating the challenges they pose, ensuring a smoother transition, and enhancing the overall success of the blockchain implementation process.

During this final stage of the BT adoption process, strategies such as putting BT into use and confirming the decision to continue using it take place. Firstly, the company must be prepared to receive innovation. In this way, strategies focusing on establishing an effective blockchain governance model, IoT blockchain integration for data reliability, and co-certification approach for blockchain transactions, as well as the other strategies mentioned in this stage, will help with the technology's integration into the organization's operations. Implementing these strategies will improve BT performance during pilot tests and PoCs, and successful results at this moment are likely to lead to widespread official approval of the technology. Furthermore, using these strategies significantly minimizes the probability that the organization would decide to discontinue using the technology after deciding to embrace it.

Understanding the barriers that will exist at each stage of the BT adoption process benefits adopting organizations in a variety of ways. We can thus emphasize that efforts that mitigate barriers during the initiation stage have a strong potential for reducing initial resistance and giving the organization with a better knowledge of the benefits of BT and how it may favorably impact its operations. In the adoption-decision stage, these strategies will assist businesses in making informed decisions, generating arguments that increase confidence in a decision to adopt BT. Finally, the strategies

used during the implementation stage enable the organization to achieve efficient implementation and optimize the benefits of adopting this technology.

5. Conclusions

5.1 Final remarks and contributions

This research conducted an SLR to examine technological barriers in BT implementation for SCs and ways to address them. Despite reviewing 155 publications, half were theoretical-conceptual studies, indicating an ongoing immaturity in research. Among the identified 16 technological barriers (Table 3), scalability and privacy issues were most frequently cited. The identification of these barriers facilitated the exploration of mitigation strategies.

Our comprehensive study on the integration of BT within SCM culminates in a framework that strategically addresses the technological barriers across the adoption stages – initiation, adoption decision and implementation – as proposed by the DOI theory. This framework not only substantiates the theoretical stages of DOI with practical mitigation strategies but also provides a structured methodology for successfully navigating the complex landscape of BT adoption. The academic implications of our work are profound, offering a significant extension to DOI theory's application in SCM and contributing a refined model to the blockchain literature. This model advances the theoretical understanding of BT adoption processes and sets a precedent for future research in this dynamic field.

The human element is a crucial observation that this study should emphasize when discussing the barriers to the adoption of BT. The adoption of BT is intensive in its relationship with individuals, even though the fact that the study only addresses technological barriers. Consequently, our research assigns the same level of significance to a barrier known as human reluctance to implement BT as it does to technological barriers. The decision maker or prospective users of the technology may be hesitant to adopt it due to the majority of the technological barriers mentioned in this study. Therefore, it is imperative to make coordinated efforts to increase awareness of these barriers, their potential mitigation strategies, and the moment in which they must be confronted.

Human reluctance can be converted into the adoption of technology through the implementation of a well-informed framework. We comprehend that successful BT implementations indicate that companies have successfully overcome and advanced adoption barriers, thereby converting human reluctance into technology acceptance.

Practically, our study serves as an important guide for industry practitioners, outlining a series of actionable strategies tailored to facilitate BT adoption at various critical junctures. Our findings underscore the importance of addressing specific challenges such as the *black-box effect, interoperability, standardization, scalability, security and data reliability*. For policymakers and standard-setting bodies, this research provides an evidential basis to support the development of frameworks and regulations that foster the adoption of BT.

5.2 Academic contributions

Our research makes significant contributions to the academic literature, particularly in the domains of BT and the DOI theory. Studies highlight the need for research that investigates the adoption of BT in the organizational and decision-making context (Frizzo-Barker *et al.*, 2020; Upadhyay, 2020), and the need for more efforts to understand the stages of the BT adoption process (Wamba and Queiroz, 2020). One of the primary academic contributions is the application and extension of DOI theory within the context of BT adoption in SCM. By mapping out the technological barriers and corresponding strategies to overcome these barriers across the various stages of BT adoption as outlined by DOI theory, our study provides a structured and nuanced understanding of the adoption process. This approach enriches the academic discourse surrounding DOI theory and extends its applicability to the burgeoning field of BT.

Furthermore, our research contributes to the burgeoning literature on blockchain by examining the specific technological barriers to its adoption within SCM. By identifying and categorizing these barriers, our study fills a significant gap in the literature, which often discusses BT broadly without delving into the specific challenges that organizations face during the adoption process. This detailed analysis advances our understanding of BT's implications for SCM and sets the stage for future research to build upon our findings and explore new dimensions of BT adoption.

Moreover, our proposed framework provides a structured lens for scholars to examine the complexities of blockchain adoption. It extends the academic dialogue by mapping out a clear trajectory of barrier mitigation strategies, offering a refined model that future research can use to explore the nuanced dynamics of BT integration across various industries.

5.3 Practical contributions

On the practical side, our research offers invaluable insights for managers and companies navigating the complexities of BT implementation within SCM. By outlining strategies to overcome the identified barriers at different phases of the adoption process, our study acts as a guide for practitioners. These strategies, grounded in both literature and empirical findings, provide a roadmap for organizations to navigate the often tumultuous journey of BT adoption.

Our proposed framework equips practitioners with a clear understanding of when and how to apply these strategies, thereby enhancing the efficiency, reliability, and security of BT within their SC operations, driving innovation and fostering a competitive edge in the marketplace. For the initiation phase, it suggests foundational actions; for the decision-making phase, it proposes evaluation criteria and commitment strategies; and for the implementation phase, it outlines execution tactics. For example, during the initiation phase, our research highlights the importance of demystifying BT to overcome the "black-box" effect, thus enabling managers to make informed decisions about its adoption. In the adoption-decision phase, we underscore the need for interoperability and standardization, offering strategies for organizations to ensure their BT initiatives are compatible with existing and future systems. Finally, in the implementation phase, our study offers solutions for addressing challenges such as scalability, security, and data reliability, which are critical for the successful integration of BT into SCM operations.

These practical contributions are not only valuable for organizations currently considering or in the process of

Table 5 Future research directions

Directions for future research	Suggested research questions
Investigating nontechnological barriers and ways to overcome them	• What are the nontechnological barriers to the adoption of BT in SCM? • How to overcome nontechnological barriers to the adoption of BT in SCM?
Understanding barriers in specific application contexts	• What are the barriers to adopting BT in healthcare? How to overcome them? • What are the barriers to BT adoption in SMEs? How to overcome them? • What are the barriers to adopting BT in governments? How to overcome them?
Exploring nontechnological barriers in the stages of the BT adoption process	• At what stages during the BT adoption process do nontechnological barriers arise?
Establishing the barriers that should be prioritized	• What is the influence relationship between the barriers? • What are the barriers that most influence others? • What are the barriers that should be prioritized?
Developing standards for BT that are globally accepted/ solving interoperability issues	• How to develop a global/supranational standard for BT solutions?
Investigating the readiness of organizations to overcome barriers	• Are organizations ready to overcome barriers related to BT adoption?
Improving communication between BT and enterprise management systems	• How to improve interoperability between blockchain and ERP? • How to improve interoperability between blockchain and electronic data interchange (EDI)?
Exploring ways to facilitate understanding of how the blockchain works	• How to facilitate understanding of how BT works? • How to open the black-box for understanding the technical fundamentals of blockchain?
Developing human resources to work with BT	• How to improve the qualification of the workforce on BT?

Source: Authors' own work

adopting BT but also for policymakers and standard-setting bodies looking to facilitate and regulate BT adoption. By providing a clear understanding of the barriers and strategies at each stage of adoption, our research empowers stakeholders across the SCM ecosystem to make informed decisions and take proactive steps toward leveraging BT's full potential.

5.4 Limitations and future research

Like all studies, this research presents some limitations. This topic presents research opportunities arising from elements not covered by our study and the literature. Despite our efforts to comprehensively analyze barriers and strategies, the evolving nature of the field may mean some elements could be absent from our discussion, so the analyzes and examples about strategies to mitigate barriers were not exhausted. Table 5 shows future research directions, accompanied by examples of possible RQs. It proposes investigating nontechnological barriers, understanding the relationship between barriers and adoption stages, analyzing barrier influence relationships, improving interoperability standards, assessing organizational readiness, integrating BT with other systems, addressing skills shortages, clarifying BT's comparative advantages and enhancing understanding and acceptance of BT among SC partners and customers.

References

Abeyratne, S. and Monfared, R. (2016), "Blockchain ready manufacturing supply chain using distributed ledger", *International Journal of Research in Engineering and Technology*, Vol. 05 No. 9, pp. 1-10, doi: 10.15623/ijret.2016.0509001.

Adel, H.M. and Younis, R.A.A. (2023), "Interplay among blockchain technology adoption strategy, e-supply chain management diffusion, entrepreneurial orientation and human resources information system in banking", *International Journal of Emerging Markets*, Vol. 18 No. 10, pp. 3588-3615, doi: 10.1108/IJOEM-02-2021-0165.

Agarwal, U., Rishiwal, V., Member, S., Sharma, G., Bokoro, P.N. and Sharma, R. (2022), "Blockchain technology for secure supply chain management: a comprehensive review", *IEEE Access*, Vol. 10, pp. 85493-85517, doi: 10.1109/ACCESS.2022.3194319.

Agi, M.A.N. and Jha, A.K. (2022), "Blockchain technology in the supply chain: an integrated theoretical perspective of organizational adoption", *International Journal of Production Economics*, Vol. 247, p. 108458, doi: 10.1016/j.ijpe.2022.108458.

Ahl, A., Yarime, M., Goto, M., Chopra, S.S., Kumar, N.M., Tanaka, K. and Sagawa, D. (2020), "Exploring blockchain for the energy transition: opportunities and challenges based on a case study in Japan", *Renewable and Sustainable Energy Reviews*, Vol. 117, p. 109488, doi: 10.1016/j.rser.2019.109488.

Ahram, T., Sargolzaei, A., Sargolzaei, S., Daniels, J. and Amaba, B. (2017), "Blockchain technology innovations", *2017 IEEE Technology & Engineering Management Conference (TEMSCON)*, IEEE, pp. 137-141, 10.1109/TEMSCON.2017.7998367.

Ajzen, I. (1991), "The theory of planned behavior", *Organizational Behavior and Human Decision Processes*, Vol. 50 No. 2, pp. 179-211, doi: 10.1016/0749-5978(91)90020-T.

Alam Khan, F., Asif, M., Ahmad, A., Alharbi, M. and Aljuaid, H. (2020), "Blockchain technology, improvement suggestions,

security challenges on smart grid and its application in healthcare for sustainable development", *Sustainable Cities and Society*, Vol. 55, p. 102018, doi: 10.1016/j.scs.2020.102018.

Al-Jaroodi, J. and Mohamed, N. (2019), "Blockchain in industries: a survey", *IEEE Access*, Vol. 7, pp. 36500-36515, doi: 10.1109/ACCESS.2019.2903554.

Alkafaji, B.K.A., Dashtbayaz, M.L. and Salehi, M. (2023), "The impact of blockchain on the quality of accounting information: an Iraqi case study", *Risks*, Vol. 11 No. 3, p. 58, doi: 10.3390/risks11030058.

Allen, D.W.E., Berg, C., Davidson, S., Novak, M. and Potts, J. (2019), "International policy coordination for blockchain supply chains", *Asia & the Pacific Policy Studies*, Vol. 6 No. 3, pp. 367-380, doi: 10.1002/app5.281.

Alrubei, S.M., Ball, E.A., Rigelsford, J.M. and Willis, C.A. (2020), "Latency and performance analyses of Real-World wireless IoT-Blockchain application", *IEEE Sensors Journal*, Vol. 20 No. 13, pp. 7372-7383, doi: 10.1109/JSEN.2020.2979031.

Alzahrani, N. and Bulusu, N. (2018), "Block-supply chain: a new anti-counterfeiting supply chain using NFC and blockchain", CRYBLOCK 2018 - Proceedings of the 1st Workshop on Cryptocurrencies and Blockchains for Distributed Systems, Part of MobiSys 2018, pp. 30-35, doi: 10.1145/3211933.3211939.

Andoni, M., Robu, V., Flynn, D., Abram, S., Geach, D., Jenkins, D. and McCallum, P. (2019), "Blockchain technology in the energy sector: a systematic review of challenges and opportunities", *Renewable and Sustainable Energy Reviews*, Vol. 100, pp. 143-174, doi: 10.1016/j.rser.2018.10.014.

Azaria, A., Ekblaw, A., Vieira, T. and Lippman, A. (2016), "MedRec: using blockchain for medical data access and permission management", Proceedings - 2016 2nd International Conference on Open and Big Data, OBD 2016, pp. 25-30, doi: 10.1109/OBD.2016.11.

Azzi, R., Chamoun, R.K. and Sokhn, M. (2019), "The power of a blockchain-based supply chain", *Computers and Industrial Engineering*, Vol. 135, pp. 582-592, doi: 10.1016/j.cie.2019.06.042.

Babich, V. and Hilary, G. (2019), "Blockchain and other distributed ledger technologies in operations", *Foundations and Trends® in Technology, Information and Operations Management*, Vol. 12 Nos 2/3, pp. 152-172, doi: 10.1561/0200000084.

Babich, V. and Hilary, G. (2020), "Distributed ledgers and operations: what operations management researchers should know about blockchain technology", *Manufacturing & Service Operations Management*, Vol. 22 No. 2, pp. 223-240, doi: 10.1287/MSOM.2018.0752.

Bag, S., Rahman, M.S., Gupta, S. and Wood, L.C. (2023), "Understanding and predicting the determinants of blockchain technology adoption and SMEs' performance", *The International Journal of Logistics Management*, Vol. 34 No. 6, pp. 1781-1807, doi: 10.1108/IJLM-01-2022-0017.

Bai, C. and Sarkis, J. (2020), "A supply chain transparency and sustainability technology appraisal model for blockchain technology", *International Journal of Production Research*, Vol. 58 No. 7, pp. 2142-2162, doi: 10.1080/00207543.2019.1708989.

Behnke, K. and Janssen, M. (2020), "Boundary conditions for traceability in food supply chains using blockchain technology", *International Journal of Information Management*, Vol. 52, p. 101969, doi: 10.1016/j.ijinfomgt.2019.05.025.

BigchainDB GmbH (2018), "BigchainDB 2.0 the blockchain database BigchainDB", available at: www.bigchaindb.com/whitepaper/bigchaindb-whitepaper.pdf

Biswas, B. and Gupta, R. (2019), "Analysis of barriers to implement blockchain in industry and service sectors", *Computers and Industrial Engineering*, Vol. 136, pp. 225-241, doi: 10.1016/j.cie.2019.07.005.

Biswas, S., Sharif, K., Li, F., Maharjan, S., Mohanty, S.P. and Wang, Y. (2020), "PoBT: a lightweight consensus algorithm for scalable IoT business blockchain", *IEEE Internet of Things Journal*, Vol. 7 No. 3, pp. 2343-2355, doi: 10.1109/JIOT.2019.2958077.

Bodkhe, U., Tanwar, S., Parekh, K., Khanpara, P., Tyagi, S., Kumar, N. and Alazab, M. (2020), "Blockchain for industry 4.0: a comprehensive review", *IEEE Access*, Vol. 8, pp. 79764-79800, doi: 10.1109/ACCESS.2020.2988579.

Boukis, A. (2019), "Exploring the implications of blockchain technology for brand–consumer relationships: a future research agenda", *Journal of Product & Brand Management*, Vol. 29 No. 3, pp. 307-320, doi: 10.1108/JPBM-03-2018-1780.

Brookbanks, M. and Parry, G.C. (2024), "The impact of industry 4.0 technologies on the resilience of established cross-border supply chains", *Supply Chain Management: An International Journal*, Vol. 29 No. 4, pp. 731-754, doi: 10.1108/SCM-07-2023-0333.

Bruce, J.D. (2018), "The mini-blockchain scheme", CRYPTONITE, available at: https://cryptonite.info/files/mbc-scheme-rev3.pdf

Casino, F., Kanakaris, V., Dasaklis, T.K., Moschuris, S., Stachtiaris, S., Pagoni, M. and Rachaniotis, N.P. (2021), "Blockchain-based food supply chain traceability: a case study in the dairy sector", *International Journal of Production Research*, Vol. 59 No. 19, pp. 5758-5770, doi: 10.1080/00207543.2020.1789238.

Chang, Y., Iakovou, E. and Shi, W. (2020), "Blockchain in global supply chains and cross border trade: a critical synthesis of the state-of-the-art, challenges and opportunities", *International Journal of Production Research*, Vol. 58 No. 7, pp. 2082-2099, doi: 10.1080/00207543.2019.1651946.

Chang, E.Y., Liao, S.W., Liu, C.T., Lin, W.C., Liao, P.W., Fu, W.K. and Mei, C.H. (2019), "DeepLinQ: distributed multi-layer ledgers for privacy-preserving data sharing", Proceedings - 2018 IEEE International Conference on Artificial Intelligence and Virtual Reality, AIVR 2018, IEEE, pp. 173-178, doi: 10.1109/AIVR.2018.00037.

Chen, S., Liu, X., Yan, J., Hu, G. and Shi, Y. (2020), "Processes, benefits, and challenges for adoption of blockchain technologies in food supply chains: a thematic analysis", *Information Systems and e-Business Management*, Vol. 19 No. 3, p. 0123456789, doi: 10.1007/s10257-020-00467-3.

Chittipaka, V., Kumar, S., Sivarajah, U., Bowden, J.L.H. and Baral, M.M. (2023), "Blockchain technology for supply chains operating in emerging markets: an empirical

examination of technology-organization-environment (TOE) framework", *Annals of Operations Research*, Vol. 327 No. 1, pp. 465-492, doi: 10.1007/s10479-022-04801-5.

Clohessy, T. and Acton, T. (2019), "Investigating the influence of organizational factors on blockchain adoption: an innovation theory perspective", *Industrial Management & Data Systems*, Vol. 119 No. 7, pp. 1457-1491, doi: 10.1108/IMDS-08-2018-0365.

Coblenz, M. (2017), "Obsidian: a safer blockchain programming language", Proceedings - 2017 IEEE/ACM 39th International Conference on Software Engineering Companion, ICSE-C 2017, pp. 97-99, doi: 10.1109/ICSE-C.2017.150.

Cole, R., Stevenson, M. and Aitken, J. (2019), "Blockchain technology: implications for operations and supply chain management", *Supply Chain Management: An International Journal*, Vol. 24 No. 4, pp. 469-483, doi: 10.1108/SCM-09-2018-0309.

Corallo, A., De Giovanni, M., Latino, M.E. and Menegoli, M. (2024), "Leveraging on technology and sustainability to innovate the supply chain: a proposal of Agri-food value chain model", *Supply Chain Management: An International Journal*, Vol. 29 No. 3, pp. 661-683, doi: 10.1108/SCM-12-2022-0484.

Croman, K., Decker, C., Eyal, I., Gencer, A.E., Juels, A., Kosba, A. and Miller, A. (2016), "On scaling decentralized blockchains", *FC 2016: Financial Cryptography and Data Security*, pp. 106-125, doi: 10.1007/978-3-662-53357-4_8.

Davis, F.D. (1989), "Perceived usefulness, perceived ease of use, and user acceptance of information technology", *MIS Quarterly*, Vol. 13 No. 3, p. 319, doi: 10.2307/249008.

De Filippi, P., Mannan, M. and Reijers, W. (2020), "Blockchain as a confidence machine: the problem of trust & challenges of governance", *Technology in Society*, Vol. 62, p. 101284, doi: 10.1016/j.techsoc.2020.101284.

Demir, M., Turetken, O. and Mashatan, A. (2020), "An enterprise transformation guide for the inevitable blockchain disruption", *Computer*, Vol. 53 No. 6, pp. 34-43, doi: 10.1109/MC.2019.2956927.

Dressel, W. (2017), "Engineering village", *The Charleston Advisor*, Vol. 19 No. 1, pp. 19-22, doi: 10.5260/chara.19.1.19.

Drljevic, N., Aranda, D.A. and Stantchev, V. (2020), "Perspectives on risks and standards that affect the requirements engineering of blockchain technology", *Computer Standards and Interfaces*, Vol. 69, p. 103409, doi: 10.1016/j.csi.2019.103409.

Dutta, P., Choi, T.M., Somani, S. and Butala, R. (2020), "Blockchain technology in supply chain operations: applications, challenges and research opportunities", *Transportation Research Part E: Logistics and Transportation Review*, Vol. 142, p. 102067, doi: 10.1016/j.tre.2020.102067.

Essien, A., Chukwukelu, G.O., Kazantsev, N. and Subramanian, N. (2023), "Unveiling the factors influencing transparency and traceability in Agri-food supply chains: an interconnected framework", *Supply Chain Management: An International Journal*, Vol. 29 No. 3, pp. 602-619, doi: 10.1108/SCM-02-2023-0083.

Eyal, I., Gencer, A.E., Sirer, E.G. and Van Renesse, R. (2016), "Bitcoin-NG: a scalable blockchain protocol", Proceedings of the 13th USENIX Symposium on Networked Systems Design and Implementation, NSDI 2016, *Santa Clara, CA, USA*, pp. 45-59.

Eyers, J. (2018), "Australia in driving seat as global blockchain standards take shape", available at: www.afr.com/technology/australia-in-driving-seat-as-global-blockchain-standards-take-shape-20180907-h151w7

Fan, K., Wang, S., Ren, Y., Li, H. and Yang, Y. (2018), "MedBlock: efficient and secure medical data sharing via blockchain", *Journal of Medical Systems*, Vol. 42 No. 8, pp. 1-11, doi: 10.1007/s10916-018-0993-7.

Farooque, M., Jain, V., Zhang, A. and Li, Z. (2020), "Fuzzy DEMATEL analysis of barriers to blockchain-based life cycle assessment in China", *Computers & Industrial Engineering*, Vol. 147, p. 106684, doi: 10.1016/j.cie.2020.106684.

Feng, Q., He, D., Zeadally, S., Khan, M.K. and Kumar, N. (2019), "A survey on privacy protection in blockchain system", *Journal of Network and Computer Applications*, Vol. 126, pp. 45-58, doi: 10.1016/j.jnca.2018.10.020.

Feng, H., Wang, X., Duan, Y., Zhang, J. and Zhang, X. (2020), "Applying blockchain technology to improve Agri-food traceability: a review of development methods, benefits and challenges", *Journal of Cleaner Production*, Vol. 260, p. 121031, doi: 10.1016/j.jclepro.2020.121031.

Fichman, R.G. (2000), "The diffusion and assimilation of information technology innovations", *Framing the Domains of IT Management: Projecting the Future through the past*, pp. 105-127, doi: 10.1.1.24.4539.

Fishbein, M. and Ajzen, I. (1977), "An introduction to theory and research", *Belief, Attitude, Intention, and Behavior, An Introduction to Theory and Research*, Addison-Wesley, Reading, MA.

Fitzhan, J.A., Cleine, J., Viriando, Y.F., Wang, G. and Sfenrianto, Y. (2020), "Opportunities and challenges blockchain-based smart contracts in Indonesia", *International Journal of Emerging Trends in Engineering Research*, Vol. 8 No. 2, pp. 251-258, doi: 10.30534/ijeter/2020/02822020.

Fosso Wamba, S., Queiroz, M.M. and Trinchera, L. (2020a), "Dynamics between blockchain adoption determinants and supply chain performance: an empirical investigation", *International Journal of Production Economics B.V*, Vol. 229, p. 107791, doi: 10.1016/j.ijpe.2020.107791.

Fosso Wamba, S., Kala Kamdjoug, J.R., Epie Bawack, R. and Keogh, J.G. (2020b), "Bitcoin, blockchain and fintech: a systematic review and case studies in the supply chain", *Production Planning and Control*, Vol. 31 Nos 2/3, pp. 115-142, doi: 10.1080/09537287.2019.1631460.

Frambach, R.T. and Schillewaert, N. (2002), "Organizational innovation adoption: a multi-level framework of determinants and opportunities for future research", *Journal of Business Research*, Vol. 55 No. 2, pp. 163-176, doi: 10.1016/S0148-2963(00)00152-1.

Francisco, K. and Swanson, D. (2018), "The supply chain has No clothes: technology adoption of blockchain for supply chain transparency", *Logistics*, Vol. 2 No. 1, p. 2, doi: 10.3390/logistics2010002.

Frizzo-Barker, J., Chow-White, P.A., Adams, P.R., Mentanko, J., Ha, D. and Green, S. (2020), "Blockchain as a disruptive technology for business: a systematic review", *International*

Journal of Information Management, Vol. 51, p. 102029, doi: 10.1016/j.ijinfomgt.2019.10.014.

Fu, Y. and Zhu, J. (2019), "Operation mechanism for g2b system based on blockchain", *Tehnicki Vjesnik*, Vol. 26, pp. 1841-1852, doi: 10.17559/TV-20190722100223.

Galvez, J.F., Mejuto, J.C. and Simal-Gandara, J. (2018), "Future challenges on the use of blockchain for food traceability analysis", *TrAC - Trends in Analytical Chemistry*, Vol. 107, pp. 222-232, doi: 10.1016/j.trac.2018.08.011.

Gavel, Y. and Iselid, L. (2008), "Web of science and scopus: a journal title overlap study", *Online Information Review*, Vol. 32 No. 1, pp. 8-21, doi: 10.1108/14684520810865958.

Ghadge, A., Bourlakis, M., Kamble, S. and Seuring, S. (2023), "Blockchain implementation in pharmaceutical supply chains: a review and conceptual framework", *International Journal of Production Research*, Vol. 61 No. 19, pp. 6633-6651, doi: 10.1080/00207543.2022.2125595.

Ghode, D., Yadav, V., Jain, R. and Soni, G. (2020a), "Adoption of blockchain in supply chain: an analysis of influencing factors", *Journal of Enterprise Information Management*, Vol. 33 No. 3, pp. 437-456, doi: 10.1108/JEIM-07-2019-0186.

Ghode, D.J., Yadav, V., Jain, R. and Soni, G. (2020b), "Blockchain adoption in the supply chain: an appraisal on challenges", *Journal of Manufacturing Technology Management*, Vol. 32 No. 1, pp. 42-62, doi: 10.1108/JMTM-11-2019-0395.

Gonczol, P., Katsikouli, P., Herskind, L. and Dragoni, N. (2020), "Blockchain implementations and use cases for supply Chains-A survey", *IEEE Access*, Vol. 8, pp. 11856-11871, doi: 10.1109/ACCESS.2020.2964880.

Gopalakrishnan, S. and Damanpour, F. (1997), "A review of innovation research in economics, sociology and technology management", *Omega*, Vol. 25 No. 1, pp. 15-28, doi: 10.1016/S0305-0483(96)00043-6.

Grigoreva, E.A., Garifova, L.F. and Polovkina, E.A. (2019), "The future of digital technology in Russia: blockchain as one of the priority directions of development", International Journal on Emerging Technologies, available at: https://dspace.kpfu.ru/xmlui/handle/net/157615

Hackius, N. and Petersen, M. (2020), "Translating high hopes into tangible benefits: how incumbents in supply chain and logistics approach blockchain", *IEEE Access*, Vol. 8, pp. 34993-35003, doi: 10.1109/ACCESS.2020.2974622.

Hald, K.S. and Kinra, A. (2019), "How the blockchain enables and constrains supply chain performance", *International Journal of Physical Distribution & Logistics Management*, Vol. 49 No. 4, pp. 376-397, doi: 10.1108/IJPDLM-02-2019-0063.

Hameed, M.A., Counsell, S. and Swift, S. (2012), "A conceptual model for the process of IT innovation adoption in organizations", *Journal of Engineering and Technology Management - JET-M B.V*, Vol. 29 No. 3, pp. 358-390, doi: 10.1016/j.jengtecman.2012.03.007.

Hamida, E., Brousmiche, K.L., Levard, H. and Thea, E. (2017), "Blockchain for enterprise: overview, opportunities and challenges", The 13th International Conference on Wireless and Mobile Communications, https://hal.archives-ouvertes.fr/hal-01591859%0Ahttps://hal.archives-ouvertes.fr/hal-01591859/document

Hang, Choi. and Kim. (2019), "A novel EMR integrity management based on a medical blockchain platform in hospital", *Electronics*, Vol. 8 No. 4, p. 467, doi: 10.3390/electronics8040467.

Haque, E.U., Shah, A., Iqbal, J., Ullah, S.S., Alroobaea, R. and Hussain, S. (2024), "A scalable blockchain based framework for efficient IoT data management using lightweight consensus", *Scientific Reports*, Vol. 14 No. 1, p. 7841, doi: 10.1038/s41598-024-58578-7.

Hofmann, F., Wurster, S., Ron, E. and Böhmecke-Schwafert, M. (2017), "The immutability concept of blockchains and benefits of early standardization", Proceedings of the 2017 ITU Kaleidoscope Academic Conference: Challenges for a Data-Driven Society, ITU K 2017, Vol. 2018-Janua, pp. 1-8, doi: 10.23919/ITU-WT.2017.8247004.

Howson, P. (2020), "Building trust and equity in marine conservation and fisheries supply chain management with blockchain", *Marine Policy*, Vol. 115, p. 103873, doi: 10.1016/j.marpol.2020.103873.

Huang, J., Kong, L., Chen, G., Wu, M.Y., Liu, X. and Zeng, P. (2019), "Towards secure industrial iot: blockchain system with credit-based consensus mechanism", *IEEE Transactions on Industrial Informatics*, Vol. 15 No. 6, pp. 3680-3689, doi: 10.1109/TII.2019.2903342.

Huang, H., Yue, Z., Peng, X., He, L., Chen, W., Dai, H.-N. and Zheng, Z. (2022), "Elastic resource allocation against imbalanced transaction assignments in Sharding-Based permissioned blockchains", *IEEE Transactions on Parallel and Distributed Systems*, Vol. 33 No. 10, pp. 2372-2385, doi: 10.1109/TPDS.2022.3141737.

Hughes, L., Dwivedi, Y.K., Misra, S.K., Rana, N.P., Raghavan, V. and Akella, V. (2019), "Blockchain research, practice and policy: applications, benefits, limitations, emerging research themes and research agenda", *International Journal of Information Management*, Vol. 49, pp. 114-129, doi: 10.1016/j.ijinfomgt.2019.02.005.

Hui, H., An, X., Wang, H., Ju, W., Yang, H., Gao, H. and Lin, F. (2019), "Survey on blockchain for internet of things", *Journal of Internet Services and Information Security*, Vol. 9 No. 2, pp. 1-30, doi: 10.22667/JISIS.2019.05.31.001.

Hussien, H.M., Yasin, S.M., Udzir, S.N.I., Zaidan, A.A. and Zaidan, B.B. (2019), "A systematic review for enabling of develop a blockchain technology in healthcare application: taxonomy, substantially analysis, motivations, challenges, recommendations and future direction", *Journal of Medical Systems*, Vol. 43 No. 10, pp. 1-35, doi: 10.1007/s10916-019-1445-8.

Iftekhar, A., Cui, X., Hassan, M. and Afzal, W. (2020), "Application of blockchain and internet of things to ensure Tamper-Proof data availability for food safety", *Journal of Food Quality*, Vol. 2020, p. 5385207, doi: 10.1155/2020/5385207.

ISO (2017), "ISO/TC 307: blockchain and distributed ledger technologies", available at: www.iso.org/committee/6266604.html#secretariat

Jamil, F., Hang, L., Kim, K.H. and Kim, D.H. (2019), "A novel medical blockchain model for drug supply chain integrity management in a smart hospital", *Electronics*

(Switzerland), Vol. 8 No. 5, pp. 1-32, doi: 10.3390/electronics8050505.

Jawaji, O., Asnani, B., Balasubramanian, S. and Poovammal, E. (2020), "Tracing products in a supply chain using blockchain", International Journal of Advanced Science and Technology, available at: https://sersc.org/journals/index.php/IJAST/article/view/13508

Jiang, Y., Wang, C., Wang, Y. and Gao, L. (2019), "A cross-chain solution to integrating multiple blockchains for IoT data management", *Sensors (Switzerland)*, Vol. 19 No. 9, pp. 1-18, doi: 10.3390/s19092042.

Jöhnk, J., Weißert, M. and Wyrtki, K. (2021), "Ready or not, AI comes— an interview study of organizational AI readiness factors", *Business & Information Systems Engineering*, Vol. 63 No. 1, pp. 5-20, doi: 10.1007/s12599-020-00676-7.

Justinia, T. (2019), "Blockchain technologies: opportunities for solving real-world problems in healthcare and biomedical sciences", *Acta Informatica Medica*, Vol. 27 No. 4, pp. 284-291, doi: 10.5455/aim.2019.27.284-291.

Kamble, S., Gunasekaran, A. and Arha, H. (2019), "Understanding the blockchain technology adoption in supply chains-Indian context", *International Journal of Production Research*, Vol. 57 No. 7, pp. 2009-2033, doi: 10.1080/00207543.2018.1518610.

Kamble, S.S., Gunasekaran, A. and Sharma, R. (2020), "Modeling the blockchain enabled traceability in agriculture supply chain", *International Journal of Information Management*, Vol. 52, p. 101967, doi: 10.1016/j.ijinfomgt.2019.05.023.

Kamilaris, A., Fonts, A. and Prenafeta-Boldύ, F.X. (2019), "The rise of blockchain technology in agriculture and food supply chains", *Trends in Food Science & Technology*, Vol. 91, pp. 640-652, doi: 10.1016/j.tifs.2019.07.034.

Khan, W.Z., Rehman, M.H., Zangoti, H.M., Afzal, M.K., Armi, N. and Salah, K. (2020), "Industrial internet of things: recent advances, enabling technologies and open challenges", *Computers & Electrical Engineering*, Vol. 81, p. 106522, doi: 10.1016/j.compeleceng.2019.106522.

Khezr, S., Moniruzzaman, M., Yassine, A. and Benlamri, R. (2019), "Blockchain technology in healthcare: a comprehensive review and directions for future research", *Applied Sciences*, Vol. 9 No. 9, p. 1736, doi: 10.3390/app9091736.

Kim, J.W. (2020), "Blockchain technology and its applications: case studies", *Journal of System and Management Sciences*, Vol. 10 No. 1, pp. 83-93, doi: 10.33168/jsms.2020.0106.

Kitchenham, B. and Charters, S.M. (2007), "Guidelines for performing systematic literature reviews in software engineering".

Klöckner, M., Kurpjuweit, S., Velu, C. and Wagner, S.M. (2020), "Does blockchain for 3D printing offer opportunities for business model innovation?", *Research-Technology Management*, Vol. 63 No. 4, pp. 18-27, doi: 10.1080/08956308.2020.1762444.

Kodym, O., Kubáč, L. and Kavka, L. (2020), "Risks associated with logistics 4.0 and their minimization using blockchain", *Open Engineering*, Vol. 10 No. 1, pp. 74-85, doi: 10.1515/eng-2020-0017.

Köhler, S. and Pizzol, M. (2020), "Technology assessment of blockchain-based technologies in the food supply chain", *Journal of Cleaner Production*, Vol. 269, p. 122193, doi: 10.1016/j.jclepro.2020.122193.

Kosba, A., Miller, A., Shi, E., Wen, Z. and Papamanthou, C. (2016), "Hawk: the blockchain model of cryptography and Privacy-Preserving smart contracts", Proceedings - 2016 IEEE Symposium on Security and Privacy, SP 2016, IEEE, pp. 839-858, doi: 10.1109/SP.2016.55.

Kouhizadeh, M., Saberi, S. and Sarkis, J. (2021), "Blockchain technology and the sustainable supply chain: theoretically exploring adoption barriers", *International Journal of Production Economics B.V*, Vol. 231, p. 107831, doi: 10.1016/j.ijpe.2020.107831.

Kouhizadeh, M., Sarkis, J. and Zhu, Q. (2019), "At the nexus of blockchain technology, the circular economy, and product deletion", *Applied Sciences (Switzerland)*, Vol. 9 No. 8, p. 1712, doi: 10.3390/app9081712.

Kouhizadeh, M., Zhu, Q. and Sarkis, J. (2020), "Blockchain and the circular economy: potential tensions and critical reflections from practice", *Production Planning & Control*, Vol. 31 Nos 11/12, pp. 950-966, doi: 10.1080/09537287.2019.1695925.

Krippendorff, K. (2019), "Content analysis an introduction to its methodology fourth edition".

Kumar Singh, R., Mishra, R., Gupta, S. and Mukherjee, A.A. (2023), "Blockchain applications for secured and resilient supply chains: a systematic literature review and future research agenda", *Computers & Industrial Engineering*, Vol. 175, p. 108854, doi: 10.1016/j.cie.2022.108854.

Kumar, A., Liu, R. and Shan, Z. (2020), "Is blockchain a silver bullet for supply chain management? Technical challenges and research opportunities", *Decision Sciences*, Vol. 51 No. 1, pp. 8-37, doi: 10.1111/deci.12396.

Kuo, T.T., Kim, H.E. and Ohno-Machado, L. (2017), "Blockchain distributed ledger technologies for biomedical and health care applications", *Journal of the American Medical Informatics Association*, Vol. 24 No. 6, pp. 1211-1220, doi: 10.1093/jamia/ocx068.

Kurpjuweit, S., Schmidt, C.G., Klöckner, M. and Wagner, S. M. (2021), "Blockchain in additive manufacturing and its impact on supply chains", *Journal of Business Logistics*, Vol. 42 No. 1, pp. 46-70, doi: 10.1111/jbl.12231.

Kuzlu, M., Pipattanasomporn, M., Gurses, L. and Rahman, S. (2019), "Performance analysis of a hyperledger fabric blockchain framework: throughput, latency and scalability", Proceedings - 2019 2nd IEEE International Conference on Blockchain, Blockchain 2019, IEEE, pp. 536-540, doi: 10.1109/Blockchain.2019.00003.

Lacity, M.C. (2018), "Addressing key challenges to making enterprise blockchain applications a reality", MIS Quarterly Executive, available at: https://aisel.aisnet.org/misqe/vol17/iss3/3/

Lakshmi, M.V.N.N. and Sricharan, Y.V.N.S. (2019), "Blockchain: single source of truth in shared services? an empirical paper on the relevance of blockchain for shared services", *International Journal of Recent Technology and Engineering*, p. 2763037619, available at: www.ijrte.org/portfolio-item/f2763037619/

Lee, J., Azamfar, M. and Singh, J. (2019), "A blockchain enabled Cyber-Physical system architecture for industry 4.0 manufacturing systems", *Manufacturing Letters, Society of*

Manufacturing Engineers (SME), Vol. 20, pp. 34-39, doi: 10.1016/j.mfglet.2019.05.003.

Leng, K., Bi, Y., Jing, L., Fu, H.C. and Van Nieuwenhuyse, I. (2018), "Research on agricultural supply chain system with double chain architecture based on blockchain technology", *Future Generation Computer Systems*, Vol. 86, pp. 641-649, doi: 10.1016/j.future.2018.04.061.

Li, K., Lee, J.-Y. and Gharehgozli, A. (2023), "Blockchain in food supply chains: a literature review and synthesis analysis of platforms, benefits and challenges", *International Journal of Production Research*, Vol. 61 No. 11, pp. 3527-3546, doi: 10.1080/00207543.2021.1970849.

Li, D., Liu, W., Deng, L. and Qin, B. (2021), "Design of multimedia blockchain privacy protection system based on distributed trusted communication", *Transactions on Emerging Telecommunications Technologies*, Vol. 32 No. 2, pp. 1-14, doi: 10.1002/ett.3938.

Lin, C.-F. (2019), "Blockchainizing food law: promises and perils of incorporating distributed ledger technologies to food safety, traceability, and sustainability governance", Food and Drug Law Journal, available at: https://papers.ssrn.com/sol3/papers.cfm?abstract_id=3387467

Lin, Y.P., Petway, J.R., Anthony, J., Mukhtar, H., Liao, S.W., Chou, C.F. and Ho, Y.F. (2017), "Blockchain: the evolutionary next step for ICT e-agriculture", *Environments*, Vol. 4 No. 3, pp. 1-13, doi: 10.3390/environments4030050.

Liu, X., Sun, S.X. and Huang, G. (2020), "Decentralized services computing paradigm for blockchain-based data governance: programmability, interoperability, and intelligence", *IEEE Transactions on Services Computing*, Vol. 13 No. 2, pp. 343-355, doi: 10.1109/TSC.2019.2951558.

Liu, D., Alahmadi, A., Ni, J., Lin, X. and Shen, X. (2019), "Anonymous reputation system for IIoT-Enabled retail marketing atop PoS blockchain", *IEEE Transactions on Industrial Informatics*, Vol. 15 No. 6, pp. 3527-3537, doi: 10.1109/TII.2019.2898900.

Lo, S.K., Xu, X., Chiam, Y.K. and Lu, Q. (2018), "Evaluating suitability of applying blockchain", Proceedings of the IEEE International Conference on Engineering of Complex Computer Systems, ICECCS, Vol. 2017, pp. 158-161, doi: 10.1109/ICECCS.2017.26.

Lokuge, S., Sedera, D., Grover, V. and Dongming, X. (2019), "Organizational readiness for digital innovation: development and empirical calibration of a construct", *Information & Management*, Vol. 56 No. 3, pp. 445-461, doi: 10.1016/j.im.2018.09.001.

Lu, H., Huang, K., Azimi, M. and Guo, L. (2019), "Blockchain technology in the oil and gas industry: a review of applications, opportunities, challenges, and risks", *IEEE Access*, Vol. 7, pp. 41426-41444, doi: 10.1109/ACCESS.2019.2907695.

McGhin, T., Choo, K.K.R., Liu, C.Z. and He, D. (2019), "Blockchain in healthcare applications: research challenges and opportunities", *Journal of Network and Computer Applications*, Vol. 135, pp. 62-75, doi: 10.1016/j.jnca.2019.02.027.

Mackey, T.K., Kuo, T.-T., Gummadi, B., Clauson, K.A., Church, G., Grishin, D. and Obbad, K. (2019), "Fit-for-purpose?' – Challenges and opportunities for applications of blockchain technology in the future of healthcare", *BMC Medicine*, Vol. 17 No. 1, p. 68, doi: 10.1186/s12916-019-1296-7.

Makridakis, S. and Christodoulou, K. (2019), "Blockchain: current challenges and future prospects/applications", *Future Internet*, Vol. 11 No. 12, p. 258, doi: 10.3390/fi11120258.

Mao, D., Hao, Z., Wang, F. and Li, H. (2019), "Novel automatic food trading system using consortium blockchain", *Arabian Journal for Science and Engineering*, Vol. 44 No. 4, pp. 3439-3455, doi: 10.1007/s13369-018-3537-z.

Mariappan, S. (2019), "Blockchain technology: disrupting the current business and governance model", *International Journal of Recent Technology and Engineering*, Vol. 8 No. 3, pp. 6285-6292, doi: 10.35940/ijrte.C5905.098319.

Matsuo, S. (2017), "How formal analysis and verification add security to blockchain-based systems", Proceedings of the 17th Conference on Formal Methods in Computer-Aided Design, FMCAD 2017, pp. 1-4, doi: 10.23919/FMCAD.2017.8102228.

Mattke, J., Hund, A., Maier, C. and Weitzel, T. (2019), "How an enterprise blockchain application in the U.S. Pharmaceuticals supply chain is saving lives", *MIS Quarterly Executive*, Vol. 18 No. 4, pp. 245-261, doi: 10.17705/2msqe.00019.

Mazlan, A.A., Daud, S.M., Sam, S.M., Abas, H., Rasid, S.Z., Yusof and M.F. (2020), "Scalability challenges in healthcare blockchain System-A systematic review", *IEEE Access*, Vol. 8, pp. 23663-23673, doi: 10.1109/ACCESS.2020.2969230.

Mazzei, D., Baldi, G., Fantoni, G., Montelisciani, G., Pitasi, A., Ricci, L. and Rizzello, L. (2020), "A blockchain tokenizer for industrial IOT trustless applications", *Future Generation Computer Systems B.V*, Vol. 105, pp. 432-445, doi: 10.1016/j.future.2019.12.020.

Mendling, J., Weber, I., V., Der Aalst, W., Brocke, J.V., Cabanillas, C., Daniel, F. and Debois, S. (2018), "Blockchains for business process management - challenges and opportunities", *ACM Transactions on Management Information Systems*, Vol. 9 No. 1, pp. 1-16, doi: 10.1145/3183367.

Meng, W., Tischhauser, E.W., Wang, Q., Wang, Y. and Han, J. (2018), "When intrusion detection meets blockchain technology: a review", *IEEE Access*, Vol. 6, pp. 10179-10188, doi: 10.1109/ACCESS.2018.2799854.

Meyer, T., Kuhn, M. and Hartmann, E. (2019), "Blockchain technology enabling the physical internet: a synergetic application framework", *Computers and Industrial Engineering*, Vol. 136, pp. 5-17, doi: 10.1016/j.cie.2019.07.006.

Min, H. (2019), "Blockchain technology for enhancing supply chain resilience", *Business Horizons*, Vol. 62 No. 1, pp. 35-45, doi: 10.1016/j.bushor.2018.08.012.

Moher, D., Shamseer, L., Clarke, M., Ghersi, D., Liberati, A., Petticrew, M. and Shekelle, P. (2015), "Preferred reporting items for systematic review and meta-analysis protocols (PRISMA-P) 2015 statement", *Systematic Reviews*, Vol. 4 No. 1, p. 1, doi: 10.1186/2046-4053-4-1.

Montecchi, M., Plangger, K. and Etter, M. (2019), "It's real, trust me! Establishing supply chain provenance using blockchain", *Business Horizons*, Vol. 62 No. 3, pp. 283-293, doi: 10.1016/j.bushor.2019.01.008.

Morkunas, V.J., Paschen, J. and Boon, E. (2019), "How blockchain technologies impact your business model",

Business Horizons, Vol. 62 No. 3, pp. 295-306, doi: 10.1016/j.bushor.2019.01.009.

Mundra, S. (2018), "Blockchain initiatives and implementation", *Information Services & Use*, Vol. 38 No. 3, pp. 187-189, doi: 10.3233/ISU-180022.

Murari, V. and Ravishankar, K.C. (2020), "Blockchain technology, use cases and content delivery network", *International Journal of Advanced Trends in Computer Science and Engineering*, Vol. 9 No. 2, pp. 1623-1627, doi: 10.30534/ijatcse/2020/109922020.

Nakamoto, S. (2008), "Bitcoin: a peer-to-peer electronic cash system", doi: 10.1108/TG-06-2020-0114.

Nath, S.D., Khayer, A., Majumder, J. and Barua, S. (2022), "Factors affecting blockchain adoption in apparel supply chains: does sustainability-oriented supplier development play a moderating role?", *Industrial Management & Data Systems*, Vol. 122 No. 5, pp. 1183-1214, doi: 10.1108/IMDS-07-2021-0466.

Olsen, T.L. and Tomlin, B. (2020), "Industry 4.0: opportunities and challenges for operations management", *Manufacturing & Service Operations Management*, Vol. 22 No. 1, pp. 113-122, doi: 10.1287/msom.2019.0796.

Otte, P., de Vos, M. and Pouwelse, J. (2020), "TrustChain: a sybil-resistant scalable blockchain", *Future Generation Computer Systems B*, Vol. 107, pp. 770-780, doi: 10.1016/j.future.2017.08.048. v

Padmavathi, U. and Rajagopalan, N. (2019), "A research on impact of blockchain in healthcare", *International Journal of Innovative Technology and Exploring Engineering*, Vol. 8 No. 9, pp. 35-40, doi: 10.35940/ijitee.I1007.0789S219.

Pally, V. and Reddy, V. (2019), "Enhancing supply chain management using blockchain technology", *International Journal of Engineering and Advanced Technology*, Vol. 8 No. 6, pp. 4657-4671, doi: 10.35940/ijeat.F9141.088619.

Panarello, A., Tapas, N., Merlino, G., Longo, F. and Puliafito, A. (2018), "Blockchain and iot integration: a systematic survey", *Sensors (Switzerland)*, Vol. 18 No. 8, p. 2575, doi: 10.3390/s18082575.

Pereira, J., Tavalaei, M.M. and Ozalp, H. (2019), "Blockchain-based platforms: decentralized infrastructures and its boundary conditions", *Technological Forecasting and Social Change*, Vol. 146, pp. 94-102, doi: 10.1016/j.techfore.2019.04.030.

Perez, M.R.L., Legaspi, J.B.C., Awat, K.A.S., Lagman, A.C., De Angel, R.M. and Adao, R.T. (2020), "Design and implementation of msha256 on blockchain using content addressable storage patterns", International Journal of Scientific and Technology Research, available at: www.ijstr.org/final-print/apr2020/Design-And-Implementation-Of-Msha256-On-Blockchain-Using-Content-Addressable-Storage-Patterns.pdf

Pierro, G.A. and Tonelli, R. (2022), "Can solana be the solution to the blockchain scalability problem?", 2022 IEEE International Conference on Software Analysis, Evolution and Reengineering (SANER), IEEE, pp. 1219-1226, doi: 10.1109/SANER53432.2022.00144.

Pirrong, C. (2019), "Will blockchain Be a big deal? Reasons for caution", *Journal of Applied Corporate Finance*, Vol. 31 No. 4, pp. 98-104, doi: 10.1111/jacf.12379.

Pólvora, A., Nascimento, S., Lourenço, J.S. and Scapolo, F. (2020), "Blockchain for industrial transformations: a forward-looking approach with multi-stakeholder engagement for policy advice", *Technological Forecasting and Social Change*, Vol. 157, p. 120091, doi: 10.1016/j.techfore.2020.120091.

Poon, J. and Dryja, T. (2016), "The bitcoin lightning network: scalable off-chain instant payments", May, available at: https://lightning.network/lightning-network-paper.pdf

Pournader, M., Shi, Y., Seuring, S. and Koh, S.C.L. (2020), "Blockchain applications in supply chains, transport and logistics: a systematic review of the literature", *International Journal of Production Research*, Vol. 58 No. 7, pp. 2063-2081, doi: 10.1080/00207543.2019.1650976.

Prasad, S., Shankar, R., Gupta, R. and Roy, S. (2018), "A TISM modeling of critical success factors of blockchain based cloud services", *Journal of Advances in Management Research*, Vol. 15 No. 4, pp. 434-456, doi: 10.1108/JAMR-03-2018-0027.

Prashar, D., Jha, N., Jha, S., Lee, Y. and Joshi, G.P. (2020), "Blockchain-Based traceability and visibility for agricultural products: a decentralized way of ensuring food safety in India", *Sustainability*, Vol. 12 No. 8, p. 3497, doi: 10.3390/su12083497.

Queiroz, M.M. and Fosso Wamba, S. (2019), "Blockchain adoption challenges in supply chain: an empirical investigation of the main drivers in India and the USA", *International Journal of Information Management*, Vol. 46, pp. 70-82, doi: 10.1016/j.ijinfomgt.2018.11.021.

Queiroz, M.M., Telles, R. and Bonilla, S.H. (2019a), "Blockchain and supply chain management integration: a systematic review of the literature", *Supply Chain Management: An International Journal*, Vol. 25 No. 2, pp. 241-254, doi: 10.1108/SCM-03-2018-0143.

Queiroz, M.M., Pereira, S.C.F., Telles, R. and Machado, M.C. (2019b), "Industry 4.0 and digital supply chain capabilities: a framework for understanding digitalisation challenges and opportunities", *Benchmarking: An International Journal*, Vol. 28 No. 5, pp. 1761-1782, doi: 10.1108/BIJ-12-2018-0435.

Rane, S.B. and Narvel, Y.A.M. (2019), "Re-designing the business organization using disruptive innovations based on blockchain-IoT integrated architecture for improving agility in future industry 4.0", *Benchmarking: An International Journal*, Vol. 28 No. 5, pp. 1883-1908, doi: 10.1108/BIJ-12-2018-0445.

Roehrs, A., da Costa, C.A. and da Rosa Righi, R. (2017), "OmniPHR: a distributed architecture model to integrate personal health records", *Journal of Biomedical Informatics Inc*, Vol. 71, pp. 70-81, doi: 10.1016/j.jbi.2017.05.012.

Rogers, E.M. (2003), *Diffusion of Innovations, Free Pass*, New York, 5th ed., Vol. 5.

Rogerson, M. and Parry, G.C. (2020), "Blockchain: case studies in food supply chain visibility", *Supply Chain Management: An International Journal*, Vol. 25 No. 5, pp. 601-614, doi: 10.1108/SCM-08-2019-0300.

Rossi, M., Mueller-Bloch, C., Thatcher, J.B. and Beck, R. (2019), "Blockchain research in information systems: current trends and an inclusive future research agenda", *Journal of the Association for Information Systems*, Vol. 20 No. 9, pp. 1388-1403, doi: 10.17705/1jais.00571.

Sa, B. and Umamakeswari, A. (2018), "Role of blockchain in the internet-of-things (iot)", *International Journal of Engineering & Technology*, Vol. 7 No. 2.24, p. 109, doi: 10.14419/ijet.v7i2.24.12011.

Saberi, S., Kouhizadeh, M. and Sarkis, J. (2019a), "Blockchains and the supply chain: findings from a broad study of practitioners", *IEEE Engineering Management Review*, Vol. 47 No. 3, pp. 95-103, doi: 10.1109/EMR.2019.2928264.

Saberi, S., Kouhizadeh, M., Sarkis, J. and Shen, L. (2019b), "Blockchain technology and its relationships to sustainable supply chain management", *International Journal of Production Research*, Vol. 57 No. 7, pp. 2117-2135, doi: 10.1080/00207543.2018.1533261.

Saihi, A., Awad, M. and Ben-Daya, M. (2023), "Quality 4.0: leveraging industry 4.0 technologies to improve quality management practices – a systematic review", *International Journal of Quality & Reliability Management*, Vol. 40 No. 2, pp. 628-650, doi: 10.1108/IJQRM-09-2021-0305.

Salman, T., Zolanvari, M., Erbad, A., Jain, R. and Samaka, M. (2019), "Security services using blockchains: a state of the art survey", *IEEE Communications Surveys and Tutorials*, Vol. 21 No. 1, pp. 858-880, doi: 10.1109/COMST.2018.2863956.

Sanka, A.I. and Cheung, R.C.C. (2019), "Efficient high performance FPGA based NoSQL caching system for blockchain scalability and throughput improvement", 26th International Conference on Systems Engineering, ICSEng 2018 - Proceedings, IEEE, No. 1, pp. 1-8, doi: 10.1109/ICSENG.2018.8638204.

Schmidt, C.G. and Wagner, S.M. (2019), "Blockchain and supply chain relations: a transaction cost theory perspective", *Journal of Purchasing and Supply Management*, Vol. 25 No. 4, p. 100552, doi: 10.1016/j.pursup.2019.100552.

Scott, I.J., de Castro Neto, M. and Pinheiro, F.L. (2023), "Bringing trust and transparency to the opaque world of waste management with blockchain: a polkadot parathread application", *Computers & Industrial Engineering*, Vol. 182, p. 109347, doi: 10.1016/j.cie.2023.109347.

Seok, B., Park, J. and Park, J.H. (2019), "A lightweight hash-based blockchain architecture for industrial IoT", *Applied Sciences (Switzerland)*, Vol. 9 No. 18, p. 3740, doi: 10.3390/app9183740.

Sheel, A. and Nath, V. (2020), "Antecedents of blockchain technology adoption intentions in the supply chain", *International Journal of Business Innovation and Research*, Vol. 21 No. 4, pp. 564-584, doi: 10.1504/IJBIR.2020.106011.

Shojaei, A., Wang, J. and Fenner, A. (2020), "Exploring the feasibility of blockchain technology as an infrastructure for improving built asset sustainability", *Built Environment Project and Asset Management*, Vol. 10 No. 2, pp. 184-199, doi: 10.1108/BEPAM-11-2018-0142.

Singh, V.K., Singh, P., Karmakar, M., Leta, J. and Mayr, P. (2021), "The journal coverage of web of science, scopus and dimensions: a comparative analysis", *Scientometrics*, Vol. 126 No. 6, pp. 5113-5142, doi: 10.1007/s11192-021-03948-5.

Srivastava, S., Bhadauria, A., Dhaneshwar, S. and Gupta, S. (2019), "Traceability and transparency in supply chain management system of pharmaceutical goods through block chain", International Journal of Scientific and Technology Research, available at: www.ijstr.org/final-print/dec2019/-Traceability-And-Transparency-In-Supply-Chain-Management-System-Of-Pharmaceutical-Goods-Through-Block-Chain.pdf

Sternberg, H.S., Hofmann, E. and Roeck, D. (2021), "The struggle is real: insights from a supply chain blockchain case", *Journal of Business Logistics*, Vol. 42 No. 1, pp. 71-87, doi: 10.1111/jbl.12240.

Stockman, H. (1948), "Communication by means of reflected power", *Proceedings of the IRE*, Vol. 36 No. 10, pp. 1196-1204, doi: 10.1109/JRPROC.1948.226245.

Sukheja, D., Indira, L., Sharma, P. and Chirgaiya, S. (2019), "Blockchain technology: a comprehensive survey", *Journal of Advanced Research in Dynamical and Control Systems*, Vol. 11 No. 9, pp. 1187-1203, doi: 10.5373/JARDCS/V11/20192690.

Sumarliah, E., Li, T., Wang, B., Khan, S.U. and Khan, S.Z. (2023), "Blockchain technology adoption in halal traceability scheme of the food supply chain: evidence from indonesian firms", *International Journal of Emerging Markets*, doi: 10.1108/IJOEM-05-2021-0678.

Swan, M. (2015), "Blockchain: blueprint for a new economy", O'Reilly Media Inc, Sebastopol.

Swink, M., Sant'Ana Gallo, I., Defee, C. and da Silva, A.L. (2024), "Supply chain visibility types and contextual characteristics: a literature-based synthesis", *Journal of Business Logistics*, Vol. 45 No. 1, doi: 10.1111/jbl.12366.

Tan, B.Q., Wang, F., Liu, J., Kang, K. and Costa, F. (2020), "A Blockchain-Based framework for green logistics in supply chains", *Sustainability*, Vol. 12 No. 11, p. 4656, doi: 10.3390/su12114656.

Tandon, A., Dhir, A., Islam, N. and Mäntymäki, M. (2020), "Blockchain in healthcare: a systematic literature review, synthesizing framework and future research agenda", *Computers in Industry*, Vol. 122, doi: 10.1016/j.compind.2020.103290.

Team Zilliqa (2017), "The ZILLIQA technical whitepaper", Zilliqa, available at: https://docs.zilliqa.com/whitepaper.pdf

Tezel, A., Papadonikolaki, E., Yitmen, I. and Hilletofth, P. (2020), "Preparing construction supply chains for blockchain technology: an investigation of its potential and future directions", *Frontiers of Engineering Management*, Vol. 7 No. 4, pp. 547-563, doi: 10.1007/s42524-020-0110-8.

Tian, F. (2016), "An Agri-food supply chain traceability system for China based on RFID & blockchain technology", *2016 13th International Conference on Service Systems and Service Management (ICSSSM)*, IEEE, pp. 1-6, doi: 10.1109/ICSSSM.2016.7538424.

Tiscini, R., Testarmata, S., Ciaburri, M. and Ferrari, E. (2020), "The blockchain as a sustainable business model innovation", *Management Decision*, Vol. 58 No. 8, pp. 1621-1642, doi: 10.1108/MD-09-2019-1281.

Tornatzky, L.G., Fleischer, M. and Chakrabarti, A.K. (1990), *The Processes of Technological Innovation*, Lexington Books, Lexington, MA.

Tranfield, D., Denyer, D. and Smart, P. (2003), "Towards a methodology for developing evidence-Informed management knowledge by means of systematic review", *British Journal of Management*, Vol. 14 No. 3, pp. 207-222, doi: 10.1111/1467-8551.00375.

Tripathi, G., Ahad, M.A. and Paiva, S. (2020), "S2HS-A blockchain based approach for smart healthcare system",

Healthcare Inc, Vol. 8 No. 1, p. 100391, doi: 10.1016/j.hjdsi.2019.100391.

Upadhyay, N. (2020), "Demystifying blockchain: a critical analysis of challenges, applications and opportunities", *International Journal of Information Management*, Vol. 54, p. 102120, doi: 10.1016/j.ijinfomgt.2020.102120.

Van Den Hooff, J., Kaashoek, M.F. and Zeldovich, N. (2014), "VerSum: verifiable computations over large public logs", Proceedings of the ACM Conference on Computer and Communications Security, pp. 1304-1316, doi: 10.1145/2660267.2660327.

van Hoek, R. (2019a), "Exploring blockchain implementation in the supply chain: learning from pioneers and RFID research", *International Journal of Operations & Production Management*, Vol. 39 Nos 6/7/8, pp. 829-859, doi: 10.1108/IJOPM-01-2019-0022.

van Hoek, R. (2019b), "Unblocking the chain – findings from an executive workshop on blockchain in the supply chain", *Supply Chain Management: An International Journal*, Vol. 25 No. 2, pp. 255-261, doi: 10.1108/SCM-11-2018-0383.

Vatankhah Barenji, A., Li, Z., Wang, W.M., Huang, G.Q. and Guerra-Zubiaga, D.A. (2020), "Blockchain-based ubiquitous manufacturing: a secure and reliable cyber-physical system", *International Journal of Production Research*, Vol. 58 No. 7, pp. 2200-2221, doi: 10.1080/00207543.2019.1680899.

Vazquez Melendez, E.I., Bergey, P. and Smith, B. (2024), "Blockchain technology for supply chain provenance: increasing supply chain efficiency and consumer trust", *Supply Chain Management: An International Journal*, Vol. 29 No. 4, pp. 706-730, doi: 10.1108/SCM-08-2023-0383.

Vilkov, A. and Tian, G. (2019), "Blockchain as a solution to the problem of illegal timber trade between Russia and China: SWOT analysis", *International Forestry Review*, Vol. 21 No. 3, pp. 385-400, doi: 10.1505/146554819827293231.

Vinay Reddy, V.P. (2019), "Enhancing supply chain management using blockchain technology", *International Journal of Engineering and Advanced Technology*, Vol. 8 No. 6, pp. 4657-4661, doi: 10.35940/ijeat.F9141.088619.

Viriyasitavat, W., Anuphaptrirong, T. and Hoonsopon, D. (2019), "When blockchain meets internet of things: characteristics, challenges, and business opportunities", *Journal of Industrial Information Integration*, Vol. 15, pp. 21-28, doi: 10.1016/j.jii.2019.05.002.

Wamba, S.F. and Queiroz, M.M. (2020), "Blockchain in the operations and supply chain management: benefits, challenges and future research opportunities", *International Journal of Information Management*, Vol. 52, p. 102064, doi: 10.1016/j.ijinfomgt.2019.102064.

Wan, P.K., Huang, L. and Holtskog, H. (2020), "Blockchain-Enabled information sharing within a supply chain: a systematic literature review", *IEEE Access*, Vol. 8, pp. 49645-49656, doi: 10.1109/ACCESS.2020.2980142.

Wang, Y., Han, J.H. and Beynon-Davies, P. (2019a), "Understanding blockchain technology for future supply chains: a systematic literature review and research agenda", *Supply Chain Management: An International Journal*, Vol. 24 No. 1, pp. 62-84, doi: 10.1108/SCM-03-2018-0148.

Wang, Y., Singgih, M., Wang, J. and Rit, M. (2019b), "Making sense of blockchain technology: how will it transform supply chains?", *International Journal of Production Economics B.V*, Vol. 211, pp. 221-236, doi: 10.1016/j.ijpe.2019.02.002.

Wang, J., Wu, P., Wang, X. and Shou, W. (2017), "The outlook of blockchain technology for construction engineering management", *Frontiers of Engineering Management*, Vol. 4 No. 1, p. 67, doi: 10.15302/J-FEM-2017006.

Webster, J. and Watson, R.T. (2002), "Analysing the past for prepare the future: writing a review", *MIS Quarterly*, Vol. 26 No. 2, pp. 13-23.

White, G.R.T. (2017), "Future applications of blockchain in business and management: a delphi study", *Strategic Change*, Vol. 26 No. 5, pp. 439-451, doi: 10.1002/jsc.2144.

Wohlin, C. (2014), "Guidelines for snowballing in systematic literature studies and a replication in software engineering", Proceedings of the 18th International Conference on Evaluation and Assessment in Software Engineering - EASE '14, ACM Press, *New York, NY*, pp. 1-10, doi: 10.1145/2601248.2601268.

Wong, L.-W., Leong, L.-Y., Hew, J.-J., Tan, G.W.-H. and Ooi, K.-B. (2020a), "Time to seize the digital evolution: adoption of blockchain in operations and supply chain management among malaysian SMEs", *International Journal of Information Management*, Vol. 52, p. 101997, doi: 10.1016/j.ijinfomgt.2019.08.005.

Wong, L.-W., Tan, G.W.-H., Lee, V.-H., Ooi, K.-B. and Sohal, A. (2020b), "Unearthing the determinants of blockchain adoption in supply chain management", *International Journal of Production Research*, Vol. 58 No. 7, pp. 2100-2123, doi: 10.1080/00207543.2020.1730463.

Xu, X., Zeng, Z., Yang, S. and Shao, H. (2020), "A novel blockchain framework for industrial IoT edge computing", *Sensors*, Vol. 20 No. 7, p. 2061, doi: 10.3390/s20072061.

Xu, X., Weber, I., Staples, M., Zhu, L., Bosch, J., Bass, L. and Pautasso, C. (2017), " "A taxonomy of Blockchain-Based systems for architecture design", Proceedings - 2017 IEEE International Conference on Software Architecture, ICSA 2017, pp. 243-252, doi: 10.1109/ICSA.2017.33.

Yadav, V.S., Singh, A.R., Raut, R.D. and Govindarajan, U.H. (2020), "Blockchain technology adoption barriers in the indian agricultural supply chain: an integrated approach", *Resources, Conservation and Recycling*, Vol. 161, p. 104877, doi: 10.1016/j.resconrec.2020.104877.

Yang, C.-S. (2019), "Maritime shipping digitalization: blockchain-based technology applications, future improvements, and intention to use", *Transportation Research Part E: Logistics and Transportation Review*, Vol. 131, pp. 108-117, doi: 10.1016/j.tre.2019.09.020.

Yang, R., Wakefield, R., Lyu, S., Jayasuriya, S., Han, F., Yi, X. and Yang, X. (2020), "Public and private blockchain in construction business process and information integration", *Automation in Construction*, Vol. 118, p. 103276, doi: 10.1016/j.autcon.2020.103276.

Yasaweerasinghelage, R., Staples, M. and Weber, I. (2017), "Predicting latency of Blockchain-Based systems using architectural modelling and simulation", Proceedings - 2017 IEEE International Conference on Software Architecture, ICSA 2017, pp. 253-256, doi: 10.1109/ICSA.2017.22.

Yli-Huumo, J., Ko, D., Choi, S., Park, S. and Smolander, K. (2016), "Where is current research on blockchain technology? - A systematic review", *Plos One*, Vol. 11 No. 10, p. e0163477, doi: 10.1371/journal.pone.0163477.

Yoo, M. and Won, Y. (2018), "A study on the transparent price tracing system in supply chain management based on blockchain", *Sustainability (Switzerland)*, Vol. 10 No. 11, p. 4037, doi: 10.3390/su10114037.

Younan, M., Houssein, E.H., Elhoseny, M. and Ali, A.A. (2020), "Challenges and recommended technologies for the industrial internet of things: a comprehensive review", *Measurement*, Vol. 151, p. 107198, doi: 10.1016/j.measurement.2019.107198.

Yu, Y., Liang, R. and Xu, J. (2019), "A scalable and extensible blockchain architecture", *IEEE International Conference on Data Mining Workshops, ICDMW*, IEEE, Vol. 2018, pp. 161-163, doi: 10.1109/ICDMW.2018.00030.

Zamani, E.D. and Giaglis, G.M. (2018), "With a little help from the miners: distributed ledger technology and market disintermediation", *Industrial Management & Data Systems*, Vol. 118 No. 3, pp. 637-652, doi: 10.1108/IMDS-05-2017-0231.

Zhang, J., Xue, N. and Huang, X. (2016), "A secure system for pervasive social Network-Based healthcare", *IEEE Access*, Vol. 4, pp. 9239-9250, doi: 10.1109/ACCESS.2016.2645904.

Zhang, P., White, J., Schmidt, D.C., Lenz, G. and Rosenbloom, S.T. (2018), "FHIRChain: applying blockchain to securely and scalably share clinical data", *Computational and Structural Biotechnology Journal*, Vol. 16, pp. 267-278, doi: 10.1016/j.csbj.2018.07.004.

Zhao, G., Liu, S., Lopez, C., Lu, H., Elgueta, S., Chen, H. and Boshkoska, B.M. (2019), "Blockchain technology in Agri-food value chain management: a synthesis of applications, challenges and future research directions", *Computers in Industry B.V*, Vol. 109, pp. 83-99, doi: 10.1016/j.compind.2019.04.002.

Zheng, X., Zhu, Y. and Si, X. (2019), "A survey on challenges and progresses in blockchain technologies: a performance and security perspective", *Applied Sciences (Switzerland)*, Vol. 9 No. 22, pp. 1-24, doi: 10.3390/app9224731.

Zheng, Z., Xie, S., Dai, H.N., Chen, X. and Wang, H. (2018), "Blockchain challenges and opportunities: a survey", *International Journal of Web and Grid Services*, Vol. 14 No. 4, pp. 352-375, doi: 10.1504/IJWGS.2018.095647.

Ziolkowski, R., Miscione, G. and Schwabe, G. (2020), "Decision problems in blockchain governance: old wine in new bottles or walking in someone else's shoes?", *Journal of Management Information Systems*, Vol. 37 No. 2, pp. 316-348, doi: 10.1080/07421222.2020.1759974.

Zwitter, A. and Hazenberg, J. (2020), "Decentralized network governance: blockchain technology and the future of regulation", *Frontiers in Blockchain*, Vol. 3, p. 12, doi: 10.3389/fbloc.2020.00012.

About the authors

Katherine Kaneda Moraes is an Assistant Professor at the Faculty of Engineering (Federal University of Grande Dourados), Dourados, Brazil. Her research focuses on Blockchain Technology, Industry 4.0, Supply Chain Management and Operations Management. She is a Production Engineering PhD candidate at the Federal University of São Carlos in São Carlos, Brazil. Katherine Kaneda Moraes is the corresponding author and can be contacted at: katherinemoraes@ufgd.edu.br

Gilberto Miller Devós Ganga is an Associate Professor at the Department of Production Engineering (Federal University of São Carlos), São Carlos, Brazil. Currently develops research in Logistics and Sustainable Supply Chain Management and emerging technologies, such as additive manufacturing (3D printing), the internet of things, autonomous vehicles, augmented reality, in the context of Supply Chain 4.0. He holds a Master's and Doctorate in Production Engineering from the School of Engineering of São Carlos (EESC), University of São Paulo (USP).

Moacir Godinho Filho is a Full Professor of Supply Chain Management, Operations Management and Decision Sciences (EM Normandie Business School), Le Havre, France, and also affiliated with Department of Materials and Production at Aalborg University (Denmark) and also the graduate program in Industrial Engineering at Federal University of São Carlos (Brazil). His research focuses on supply chain management, operations management, sustainable operations, industry 4.0, production planning and control, lean manufacturing and Quick Response Manufacturing. He has a PhD in Industrial Engineering from the Federal University of São Carlos – Brazil, awarded in 2004. He was also a visiting scholar at the University of Wisconsin in Madison (USA), North Carolina State University (USA) and Northwestern University (USA).

Luis Antonio Santa-Eulalia is an Associate Professor at the Department of Information Systems and Quantitative Methods in Business (Business School - Université de Sherbrooke), Sherbrooke, Canada. His research focuses on Operations management, Management of value-creation networks, Information system, Modeling and simulation of systems, Industry 4.0 and digital transformation and Sustainable management. He holds a PhD, MSc and a degree in Production Engineering, respectively, from Université Laval (Canada), University of São Paulo (Brazil) and Federal University of São Carlos (Brazil).

Guilherme Luz Tortorella is an Associate Professor of the Department of Systems and Production Engineering of the Universidade Federal de Santa Catarina, Brazil. He is the Head of Research of the Productivity and Continuous Improvement Lab and the Editor-in-Chief of the *Journal of Lean Systems*. He is one of the founders of the Brazilian Conference on Lean Systems and has more than 18 years with practical and academic experience with manufacturing and operations management.

For instructions on how to order reprints of this article, please visit our website:
www.emeraldgrouppublishing.com/licensing/reprints.htm
Or contact us for further details: **permissions@emeraldinsight.com**